

Discretization Invariant and Mesh-Free Operator Learning for Nuclear Reactor Modeling

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Discretization Invariant and Mesh-Free Operator Learning for Nuclear Reactor Modeling

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Abstract

Partial differential equations (PDEs) are central to scientific computing and engineering, but high-fidelity numerical simulations are often computationally expensive. Neural operators have recently emerged as a promising approach for surrogate modeling by learning mappings between function spaces rather than fixed-dimensional vectors.

A key challenge for such models is discretization invariance: the ability to generalize to spatial resolutions unseen during training. This thesis considers this property for Fourier- and convolution-based neural operators on uniform Cartesian grids, and for geometry-aware neural operators on non-uniform point cloud domains.

On uniform grids, the Fourier Neural Operator (FNO) and the Convolutional Neural Operator (CNO) are evaluated on Navier–Stokes and Darcy flow benchmarks through resolution-transfer experiments. The results show that CNO exhibits substantially stronger robustness across resolutions, while FNO is more sensitive to discretization mismatch due to its reliance on a fixed set of Fourier modes. An ablation study shows that this sensitivity is primarily caused by the dependence on input resolution. Introducing resampling onto a fixed internal grid reduces this sensitivity by nearly two orders of magnitude.

For non-uniform meshes, the geometry-aware Fourier Neural Operator (Geo-FNO) is evaluated on incompressible flow past a cylinder and on a high-temperature gas-cooled reactor (HTGR) temperature dataset. The results show that Geo-FNO produces accurate predictions on irregular point clouds, while also highlighting that discretization invariance in the mesh-free setting must be interpreted as robustness to changes in sampling geometry rather than resolution transfer.

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Chapter 1

Introduction

Partial differential equations (PDEs) are central to scientific computing and engineering because they describe how physical systems evolve. Fluid flow, heat transfer, structural deformation, neutron transport, and porous-media flow can all be formulated through PDEs. In nuclear engineering in particular, reactor analysis depends heavily on solving large coupled PDE systems describing neutronics, thermohydraulics, heat conduction, and structural behavior.

Classical numerical methods such as finite difference, finite element, and finite volume methods provide accurate solutions to these problems, but they are often computationally expensive. High-fidelity simulations may require fine spatial discretizations, long transient integrations, or repeated solves under varying parameters. This becomes particularly challenging in applications such as optimization, uncertainty quantification, design studies, and real-time decision support, where the same governing equations must be solved many times under different conditions.

This motivates the use of learning-based surrogate models; machine learning models trained to emulate the behavior of expensive numerical solvers while producing predictions at orders of magnitude lower computational cost [9]. Once trained, such models can provide rapid evaluations of complex physical systems, making them attractive for engineering workflows where repeated PDE solves are the primary computational bottleneck.

However, standard machine learning models are typically formulated as mappings between finite-dimensional vectors. A conventional neural network learns a mapping of the form

$$x \mapsto y,$$

where both the input and output are fixed-dimensional vectors. PDE problems, however, are naturally function-to-function problems. The input may be an initial condition, boundary condition, forcing term, or material coefficient field, while the output is itself a function such as a pressure field, temperature distribution, or velocity field. The object being learned is therefore not a vector-valued map, but an operator between function spaces.

Neural operators were introduced to address this setting directly. Rather than learning a mapping between discretized vectors, they aim to approximate the underlying continuous solution operator of a PDE. Instead of learning a map tied to one specific discretization, the

model approximates the operator itself, allowing the same learned model to be evaluated across multiple discretizations of the same underlying problem. This property is one of the main distinctions between neural operators and standard surrogate models.

This property is commonly referred to as discretization invariance. Informally, a discretization-invariant model should produce consistent predictions when the same underlying function is represented on different discretizations, provided those discretizations sufficiently resolve the relevant physics. In this sense, the model depends on the underlying continuous object rather than on the particular grid used to represent it.

Discretization invariance is one of the central criteria used to evaluate neural operators in practice. In this thesis, it is examined directly on uniform Cartesian grids through resolution-transfer experiments, and later discussed in the context of non-uniform meshes and point clouds, where the notion of discretization is less straightforward and does not admit the same type of evaluation.

This motivates the two-part structure of this thesis. The first part investigates discretization invariance on uniform Cartesian grids using two important neural operator architectures: the Fourier Neural Operator (FNO) [15] and the Convolutional Neural Operator (CNO)[19]. These models are evaluated on Navier–Stokes and Darcy flow benchmarks to study how architectural choices affect robustness under resolution transfer.

The second part considers mesh-free operator learning on irregular domains using the geometry-aware Fourier Neural Operator (Geo-FNO) [16]. The goal here is to evaluate whether operator learning can be performed directly on non-uniform meshes while preserving physically meaningful predictions. This is studied on both a standard fluid benchmark involving incompressible flow past a cylinder and a high-temperature gas-cooled reactor (HTGR) temperature dataset.

In addition to evaluating these architectures, this thesis also investigates practical implementation issues that affect operator learning on irregular domains. In particular, a bug in the original Geo-FNO [25] implementation was identified and corrected, removing a persistent seam artifact that appeared across all tested datasets and significantly improving predictive accuracy. The proposed fix was later incorporated into the official implementation.

The central research question investigated in this thesis is:

- To what extent can Fourier- and convolution-based neural operator methods provide discretization-robust surrogate models for PDE-based data on both uniform grids and non-uniform point clouds?

This is further divided into the following sub-questions:

- To what extent do FNO and CNO generalize across grid resolutions on uniform-grid PDE benchmarks?
- How effectively does Geo-FNO enable Fourier-based operator learning on non-uniform point-cloud data?
- What limitations arise when applying Geo-FNO to reactor-relevant data defined on irregular meshes?

The main contributions of this thesis are threefold. First, it provides a systematic comparison of FNO and CNO under resolution transfer, showing how spectral truncation, fixed internal resampling, and aliasing control affect discretization robustness. Second, it evaluates Geo-FNO on irregular-domain problems and clarifies how discretization invariance should be interpreted in the mesh-free setting. Third, it demonstrates the applicability of neural operators to reactor-relevant non-uniform data while identifying both practical limitations of the current datasets and implementation issues that arise in real mesh-free applications.

The remainder of this thesis is structured as follows. Chapter 2 introduces the theoretical background of operator learning, discretization invariance, and the neural operator architectures studied in this work. Chapter 3 describes the datasets, training procedures, and experimental design. Chapter 4 presents the uniform-grid experiments and discusses the behavior of FNO and CNO under resolution transfer. Chapter 5 presents the mesh-free experiments with Geo-FNO on irregular domains and discusses their implications for discretization invariance on non-uniform meshes. Finally, Chapter 6 summarizes the conclusions of the thesis and outlines directions for future work.

Chapter 2

Background & Theory

This chapter introduces the theoretical background needed for the remainder of the thesis. It first places operator learning in context with more classical learning setups and then discusses the neural operator architectures that are central to this work.

2.1 Operator Learning

Operator learning refers to the use of neural networks to approximate mappings between function spaces, rather than between finite-dimensional vectors as is done for classical neural network approaches [14]. This setting arises naturally in scientific computing, where many problems can be formulated in terms of operators acting on functions.

A particularly important example is the solution operator of a parametric partial differential equation (PDE). Consider a PDE of the form:

$$\mathcal{N}(a, u)(x) = f(x), \quad x \in \Omega, \quad u(x) = 0, \quad x \in \partial\Omega, \quad (2.1)$$

where $\Omega \subset \mathbb{R}^d$ is a bounded domain, $\mathcal{N}$ is a differential operator, a is an input function, f is an external term like a forcing term, and u is the solution.

The corresponding solution operator is defined as:

$$\mathcal{G}(a) = u, \quad (2.2)$$

which maps an input function a to the solution u . The goal of operator learning is to approximate this mapping directly using surrogate modeling.

Let $\mathcal{A}$ and $\mathcal{U}$ denote appropriate function spaces for the input and output, respectively. Given a dataset of input-output pairs:

$$\{(a_j, u_j)\}_{j=1}^{N_{\text{data}}}, \quad u_j = \mathcal{G}(a_j), \quad (2.3)$$

where the inputs a_j are sampled from a distribution μ over $\mathcal{A}$, the objective is to learn a parametric approximation $\mathcal{G}_\theta : \mathcal{A} \rightarrow \mathcal{U}$ by minimizing the expected error

$$\mathbb{E}_{a \sim \mu} [\|\mathcal{G}(a) - \mathcal{G}_\theta(a)\|_{\mathcal{U}}^2], \quad (2.4)$$

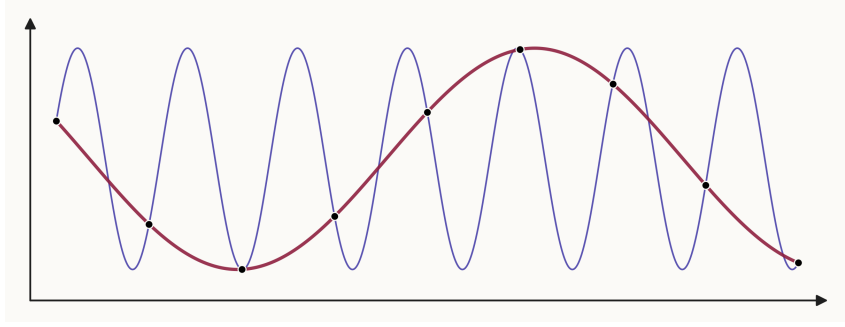


Figure 2.1: An illustration of aliasing. The blue curve represents a high-frequency signal, while the red curve is a lower-frequency function that passes through the same sampled points (black dots). Since both functions agree at the sampling locations, the discrete representation cannot distinguish between them.

which in practice is approximated by the empirical loss over the dataset.

A key distinction from standard learning problems is that the objects of interest are functions rather than finite-dimensional vectors, even though in practice they are only observed through discrete samples. Consequently, the learning problem is formulated at the level of a continuous operator, while training is carried out on its discretized representations. This makes it essential to understand what information about a function is preserved under discretization.

2.1.1 Bandlimitedness and Continuous–Discrete Equivalence

A central question in operator learning is what information about a continuous function is preserved when it is represented through discrete samples. Sampling theory provides a framework for answering this question, by characterizing when a function can be uniquely reconstructed from its values on a discrete grid.

Let $f \in L^2(\mathbb{R})$ be a square-integrable function with Fourier transform $\hat{f}$. The function is said to be *bandlimited* with bandlimit $\Omega > 0$ if its Fourier transform is supported only on a finite interval:

$$\text{supp}(\hat{f}) \subset [-\Omega, \Omega]. \quad (2.5)$$

The Whittaker-Shannon sampling theorem states that a bandlimited function can be exactly reconstructed from its values on a uniform grid, provided the sampling rate is sufficiently high [18]. Specifically, if the function is sampled at points $\{f(nT)\}_{n \in \mathbb{Z}}$, then exact reconstruction is possible if

$$\frac{1}{T} \geq 2\Omega, \quad (2.6)$$

where $1/T$ is the sampling frequency and 2Ω is called the Nyquist rate.

Under this condition, the function can be reconstructed via the interpolation formula

$$f(x) = \sum_{n \in \mathbb{Z}} f(nT) \text{sinc}\left(\frac{x - nT}{T}\right), \quad (2.7)$$

where $\text{sinc}(x) = \frac{\sin(\pi x)}{\pi x}$.

This result establishes a form of continuous-discrete equivalence: if a function is bandlimited and sampled above the Nyquist rate, then its discrete samples contain all the information required to reconstruct the original function. In this case, the discretization is effectively lossless.

However, if the sampling rate is below the Nyquist rate, this equivalence breaks down. In this case, different continuous functions can produce exactly the same set of discrete samples, meaning that the original function can no longer be uniquely reconstructed from the sampled data.

This phenomenon is known as aliasing. Intuitively, aliasing occurs when oscillations in the function are too rapid relative to the spacing between sample points. If the grid is too coarse, the samples cannot resolve the true high-frequency behavior, and the function appears to vary more slowly than it actually does.

Figure 2.1 illustrates this effect. The blue curve represents a high-frequency signal, while the red curve is a much lower-frequency function passing through the same sampled points. Since both functions agree exactly at the sampling locations, the discrete representation cannot distinguish between them. From the perspective of the sampled data, the high-frequency signal is therefore indistinguishable from its lower-frequency alias.

From a Fourier perspective, a sampled grid can only represent frequencies up to the Nyquist limit

$$\omega_N = \frac{\pi}{T}, \quad (2.8)$$

where T is the spacing between sample points. Frequencies satisfying

$$|\omega| \leq \omega_N \quad (2.9)$$

can be represented uniquely, while frequencies above this limit cannot.

Consider a sinusoidal component

$$f(x) = \sin(\omega x). \quad (2.10)$$

Sampling this function at points $x_n = nT$ gives

$$f(nT) = \sin(\omega nT). \quad (2.11)$$

If $\omega > \omega_N$, there exists a lower frequency $\omega_{\text{alias}} \leq \omega_N$ such that

$$\sin(\omega nT) = \sin(\omega_{\text{alias}} nT) \quad (2.12)$$

for all sample points n . A common expression for this aliased frequency is

$$\omega_{\text{alias}} = |\omega - 2k\omega_N|, \quad (2.13)$$

for some integer k chosen so that

$$0 \leq \omega_{\text{alias}} \leq \omega_N. \quad (2.14)$$

This shows that frequencies above the Nyquist limit do not simply disappear. Instead, they are systematically and deterministically mapped onto lower frequencies that the grid can represent. Aliasing is therefore not random noise, but a predictable distortion of the frequency content.

The consequence is that information above the Nyquist limit is irreversibly lost. Once aliasing has occurred, the sampled data contains a distorted representation of the original function, where unresolved high-frequency behavior is interpreted as physically different low-frequency structure. No learning algorithm can reconstruct frequency content that is absent from the discretized data, regardless of model complexity or architecture [1].

2.1.2 Discretization Invariance

The discussion in Section 2.1.1 shows that learning from discretized data is fundamentally constrained by the information preserved under sampling. Even if a model is intended to approximate a continuous operator, it is trained and evaluated on finite representations of functions. This raises the question of whether the learned model depends only on the underlying function, or also on the particular discretization used to represent it.

This property is commonly referred to as discretization invariance. Informally, a discretization-invariant model should produce consistent predictions when the same underlying function is represented on different grids, provided those discretizations sufficiently resolve the relevant features of the problem.

In practice, discretization invariance is most naturally interpreted for functions represented on uniform Cartesian grids, where resolution can be clearly defined through grid spacing. In this setting, it refers to the ability of a model trained at one resolution to make accurate predictions at other resolutions of the same underlying problem.

A widely used definition of neural operators is given by Kovachki et al. [14], who state that a neural operator should satisfy three key properties:

1. **Discretization flexibility:** the model can accept inputs defined on arbitrary discretizations of the domain.
2. **Pointwise evaluation:** the model can be evaluated at arbitrary output locations.
3. **Continuum consistency:** as the discretization is refined, the learned operator converges to the underlying continuous operator being approximated.

These properties formalize the idea that a neural operator should approximate a mapping between functions, rather than a mapping between specific finite-dimensional representations. In particular, the learned operator should depend on the underlying continuous object and not on the resolution, ordering, or structure of the grid used to represent it.

Since training data is always discretized, a model is never directly exposed to the continuous operator itself, but only to finite samples of its input and output functions. If refining the discretization changes the behavior of the model rather than improving the approximation of the same underlying mapping, the model becomes tied to the representation used during training rather than to the operator itself. As a result, performance may degrade

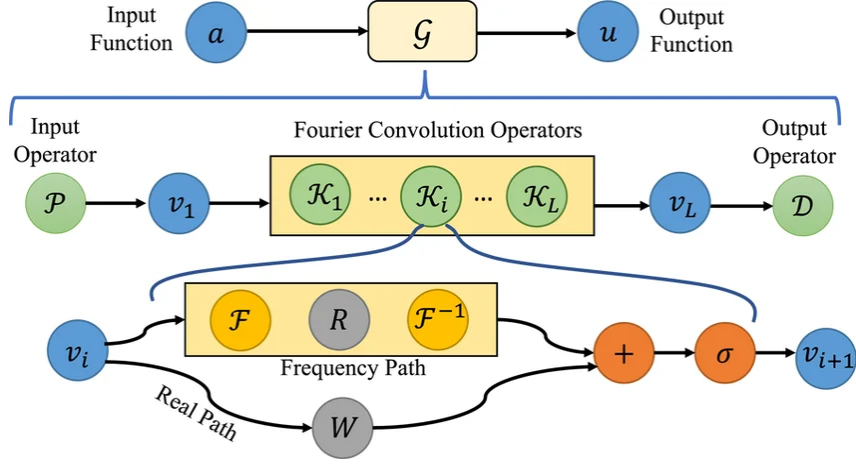


Figure 2.2: The architecture of the Fourier Neural Operator (FNO), showing the lifting layer, repeated Fourier layers consisting of spectral convolution and pointwise channel mixing, and the final projection layer. [12].

significantly when evaluated on different resolutions or different discretizations of the same problem. This phenomenon is commonly referred to as *discretization mismatch* [8].

Discretization invariance is therefore a central criterion for evaluating neural operator architectures. In this thesis, it is used both as a theoretical concept and as an empirical measure for comparing models across changes in resolution.

2.2 Fourier Neural Operator – FNO

The Fourier Neural Operator (FNO), introduced by Li et al. [15], was the first major neural operator architecture to demonstrate strong performance on large-scale PDE learning problems and remains one of the most influential models in operator learning. Its success established neural operators as a practical alternative to classical numerical solvers and motivated much of the subsequent development in the field [7], including architectures such as the Convolutional Neural Operator (CNO) and Geo-FNO.

The central idea of the FNO is to learn the solution operator in Fourier space. The model performs global convolution by transforming the input field into its Fourier representation, applying learned transformations to the Fourier coefficients, and transforming the result back to the spatial domain.

Figure 2.2 illustrates the architecture of the FNO. The model approximates the solution operator

$$\mathcal{G} : a \mapsto u, \quad (2.15)$$

where a is the input function and u is the corresponding solution field. The full network consists of three main components: an input lifting operator, a sequence of Fourier layers, and an output projection operator.

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The lifting operator maps the input field into a higher-dimensional latent representation,

$$\mathcal{P} : a(x) \rightarrow v_1(x), \quad (2.16)$$

where $v_1(x)$ contains a larger number of feature channels than the original physical field. This allows the subsequent Fourier layers to operate in a richer latent space rather than directly on the low-dimensional physical variables.

The core of the architecture is the Fourier layer. Given an intermediate representation $v_l(x)$, each layer updates the features according to

$$v_{l+1}(x) = \sigma(Wv_l(x) + \mathcal{K}(v_l)(x)), \quad (2.17)$$

where $\mathcal{K}$ is the spectral convolution operator, W is a pointwise linear transformation, and σ is a nonlinear activation function.

The spectral convolution operator is implemented by first applying a Fourier transform to the input features, followed by truncation to a fixed number of low-frequency Fourier modes. Learned complex-valued weights are then applied directly to these retained coefficients in frequency space. Finally, an inverse Fourier transform maps the result back to the spatial domain,

$$\mathcal{K}(v_l) = \mathcal{F}^{-1}(R \cdot \mathcal{F}(v_l)), \quad (2.18)$$

where R denotes the learned complex weights acting on the retained modes.

This frequency-space formulation is what distinguishes the FNO from standard convolutional architectures. Instead of learning kernels tied to a fixed spatial stencil, the model learns transformations of Fourier modes, which correspond to global basis functions spanning the entire domain.

Alongside the spectral path, the term $Wv_l(x)$ provides a local pointwise linear transformation in physical space. This is a channel-mixing operator analogous to a 1×1 convolution. It allows the model to incorporate local, pointwise corrections that cannot be represented by the truncated set of retained Fourier modes. Note that although W and $\mathcal{K}$ are linear operations, the nonlinearity σ allows the model to learn nonlinearities.

After several Fourier layers, the projection operator maps the final latent representation back to the physical output space,

$$\mathcal{D} : v_L(x) \rightarrow u(x), \quad (2.19)$$

producing the predicted solution field.

2.2.1 Strengths

One of the main strengths of the FNO is its ability to capture global dependencies efficiently. Since Fourier modes are global basis functions, modifying a single coefficient influences the solution across the entire domain. This gives the model a global receptive field in every layer, making it particularly well suited for PDEs whose solution operators exhibit long-range interactions.

A second major advantage is computational efficiency. Directly learning an integral operator in physical space would require pairwise interactions between all spatial locations,

leading to quadratic complexity in the number of grid points. By performing convolution through the Fast Fourier Transform (FFT), the FNO reduces this cost to

$$O(n \log n), \quad (2.20)$$

which makes large-scale operator learning computationally feasible.

The use of Fourier coefficients also provides a degree of resolution transfer across uniform Cartesian grids. Since the learned parameters act on modes rather than individual grid points, the same operator can be applied to different uniform resolutions without retraining, provided the relevant frequency content is sufficiently resolved. This makes the FNO significantly more flexible than standard convolutional networks, which are often strongly tied to the discretization used during training.

2.2.2 Weaknesses

The primary limitation of the FNO is its reliance on spectral truncation. In each Fourier layer, only a fixed number of low-frequency modes are retained, while the remaining higher-frequency coefficients are discarded. This introduces an explicit architectural bandlimit determined by the number of retained modes.

This should be distinguished from the sampling bandlimit discussed in Section 2.1.1. There, the Nyquist limit was imposed by the resolution of the data itself; frequencies above that limit could not be represented in the discretized input. In the FNO, an additional bandlimit is imposed by the model architecture, even if higher-frequency information is present in the data. The network therefore discards information that may in principle be available.

As discussed in the context of aliasing, unresolved high-frequency modes do not simply vanish. Instead, they are mapped onto lower-frequency components during discrete sampling, a phenomenon known as folding. Frequencies above the Nyquist limit become indistinguishable from lower frequencies, leading to a distorted representation of the underlying function. This issue becomes particularly severe when the target operator depends strongly on localized or high-frequency features, such as sharp gradients, discontinuities, or fine-scale geometric effects. In such cases, the low-mode bias of the FNO can lead to poor generalization across resolutions and unstable behavior under discretization changes.

A second important limitation is the reliance on uniform Cartesian grids. The standard FFT assumes regularly spaced points with consistent ordering, which makes the original FNO naturally suited for structured meshes but difficult to apply directly to irregular geometries or unstructured point clouds.

Finally, although the FNO is often described as discretization invariant, this property only holds approximately and within the setting of uniform grids. Changes in resolution can still lead to discretization mismatch, particularly when the effective frequency content of the problem changes across discretizations. This limitation motivates both the development of the CNO, which addresses representation consistency more explicitly, and Geo-FNO, which extends Fourier-based operator learning to irregular domains.

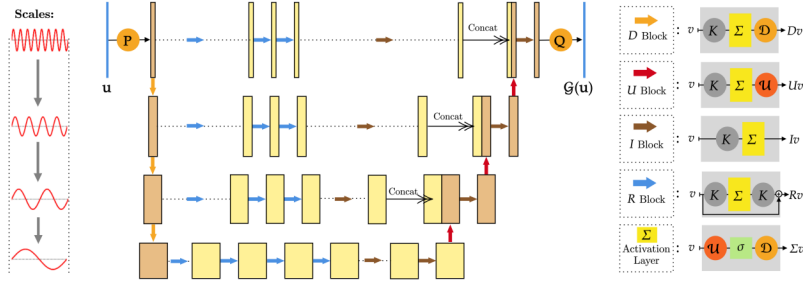


Figure 2.3: The architecture of the Convolutional Neural Operator (CNO) [19], showing the lifting operator P , projection operator Q , and the central U-shaped neural operator structure. The network combines downsampling (D -blocks), upsampling (U -blocks), residual blocks (R -blocks), and invariant blocks (I -blocks) across multiple spatial scales, with skip connections and concatenation used to preserve multiscale information. Anti-aliased downsampling and upsampling ensure resolution-invariant feature propagation across discretizations.

2.3 Convolutional Neural Operator – CNO

2.3.1 Architecture

The Convolutional Neural Operator (CNO), introduced by Raonic et al. [19], extends the classical U-Net architecture to the operator learning setting. U-Nets and related convolutional neural networks have been highly successful in fields such as image processing and segmentation due to their ability to efficiently capture local spatial structure through convolutional kernels [20]. The CNO adopts this encoder–decoder structure while introducing several modifications aimed at satisfying the discretization invariance requirements discussed in Section 2.1.2.

Unlike the Fourier Neural Operator, which learns global interactions through Fourier modes, the CNO is built on standard convolutions in physical space. A convolutional kernel acts on a local neighborhood of points, producing a local operator rather than a global spectral one. This makes the model structurally similar to standard convolutional neural networks, but introduces an important challenge for operator learning: the physical meaning of a fixed kernel depends on the discretization.

For example, a 3×3 kernel applied on a coarse grid covers a much larger physical region than the same kernel on a fine grid. As a result, a standard CNN or U-Net is not naturally discretization invariant, since changing the resolution changes the effective operator being learned.

To address this, the CNO uses a multi-scale encoder–decoder architecture together with an initial resampling step onto a fixed internal resolution. Figure 2.3 illustrates the overall structure. After the initial resampling, the input is passed through a lifting layer, which maps the physical input field into a higher-dimensional feature space. This serves the same purpose as in the FNO, allowing the operator to be learned in a richer latent representation.

The lifted representation then enters the encoder, where it passes through a sequence of convolutional blocks. Each block consists of a convolution followed by a nonlinear activation. As the representation moves deeper into the encoder, the spatial resolution is progressively reduced through interpolation-based downsampling, while the number of feature channels is increased. Although the kernel size remains fixed, applying it at multiple resolutions allows the network to process information across different spatial scales.

At the coarsest level, the decoder progressively reconstructs the solution by upsampling the representation back to higher resolutions. After each upsampling step, another convolutional block is applied. Skip connections link encoder and decoder stages at matching resolutions, allowing fine-scale information from earlier layers to be reintroduced during reconstruction.

A key modification compared to a standard U-Net is the use of resampling onto a fixed internal grid before the main operator learning begins. Regardless of the input resolution, the model first interpolates the input onto the same internal discretization. This ensures that the convolutional kernels always act over the same physical scale inside the network.

This design also connects directly to the bandlimitedness discussion in Section 2.1.1. By choosing a fixed internal resolution, the CNO imposes an explicit bandlimit on the learned representation. Frequencies beyond this scale are removed during resampling, so the model always operates on a controlled frequency range.

A second major modification is the replacement of standard pointwise nonlinearities with the upsample–nonlinearity–downsample (UND) block. A standard pointwise activation applied directly on a finite grid is not representation-equivalent: even if the input is bandlimited, the nonlinearity generates new higher-frequency components that may lie above the Nyquist limit. These unresolved frequencies alias into lower modes, making the discrete nonlinear operator inconsistent with the corresponding continuous operator and violating the continuum consistency discussed in Section 2.1.2.

To address this, the CNO first upsamples the representation before applying the nonlinearity, creating spectral room for the newly generated frequencies to exist without immediately aliasing into the resolved modes. After the activation is applied, the representation is downsampled back to the original resolution using a low-pass filtering operation that removes the unresolved high-frequency content. Ideally, this filtering would be performed using the sinc interpolation filter implied by the Whittaker–Shannon sampling theorem, which provides exact reconstruction for bandlimited functions [18]. In practice, this is approximated by a finite-support low-pass blur filter for computational efficiency, since exact sinc filtering is non-local and expensive to evaluate.

After the decoder stage, a projection layer maps the final latent representation back to the physical output space, producing the predicted solution field. Together, the fixed-grid resampling and the UND block are the central modifications that distinguish the CNO from a standard U-Net and make it suitable for operator learning across multiple discretizations.

2.3.2 Strengths

A major strength of the CNO is its strong local inductive bias. Since the model is built on standard convolutions in physical space, it naturally captures local interactions where the

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state at a point depends primarily on its immediate neighborhood. This is particularly well suited for many physical systems such as advection, diffusion, heat transfer, and turbulence-dominated flows, where local spatial structure plays a dominant role.

The encoder–decoder structure further allows the model to capture interactions across multiple physical scales. Coarse-resolution layers represent large-scale structures and long-range behavior, while fine-resolution layers preserve localized detail through skip connections. This is especially important for multiscale systems such as Navier–Stokes flow, where energy is transferred across a wide range of length scales and both global and local behavior must be represented simultaneously.

Another important advantage is robustness across resolutions. Because all inputs are first resampled onto the same internal grid, the model avoids much of the discretization dependence seen in standard convolutional architectures. Provided that the chosen internal resolution is sufficient to resolve the relevant physics, this often leads to more stable generalization across discretizations than methods that rely directly on the input resolution.

The explicit treatment of bandlimiting also provides an important advantage compared to architectures where truncation occurs only implicitly. By controlling both resampling and nonlinear aliasing through the fixed internal grid and the UND block, the CNO avoids many of the spectral artifacts associated with unresolved high-frequency content. This improves continuum consistency and makes the model much closer to satisfying the definition of a true neural operator discussed in Section 2.1.2 and in [1].

2.3.3 Weaknesses

The primary weakness of the CNO is its computational cost. Compared to the FNO, which leverages the Fast Fourier Transform to achieve $O(n \log n)$ complexity, the CNO relies on repeated convolutional operations across multiple resolutions. The encoder–decoder structure, skip connections, repeated interpolation steps, and UND blocks all increase both training time and memory usage, making the model significantly more expensive to train and evaluate.

A second limitation is the dependence on the chosen internal resolution. The resampling step improves consistency by enforcing a fixed internal representation, but it also introduces a strong modeling assumption: namely, that this chosen resolution is sufficient to capture all physically relevant features of the problem. If the internal grid is too coarse, important high-frequency information may be removed before learning even begins. In this case, the model becomes stable but systematically under-resolved.

This explicit truncation of high-frequency information is therefore both a strength and a weakness. While it improves robustness and discretization consistency, it can reduce accuracy for problems where sharp gradients, discontinuities, or fine-scale geometric details play a dominant role. In such cases, the CNO may sacrifice important local information in exchange for stability.

Another practical limitation is the reliance on interpolation during resampling. Both the initial projection onto the fixed internal grid and the repeated upsampling and downsampling operations in the UND blocks introduce approximation errors. Ideally, these operations would be performed using exact sinc interpolation for bandlimited functions, but in practice

finite-support approximations are used for computational efficiency, which can introduce errors.

Finally, while the CNO improves discretization invariance substantially compared to standard U-Nets and often compared to the FNO, it is still not fully resolution-independent in practice. Its behavior remains tied to the chosen internal discretization, meaning that discretization robustness depends strongly on how well this internal representation matches the true frequency content of the underlying problem.

2.4 Mesh-free Operator Learning and Geo-FNO

2.4.1 Motivation for Irregular Domains

The neural operators introduced in Sections 2.2 and 2.3 assume that the input and output fields are represented on uniform Cartesian grids. While this assumption is convenient for both Fourier-based and convolutional architectures, it is often restrictive for scientific and engineering problems where the natural discretization of the domain is non-uniform.

A uniform grid can be an inefficient representation when the solution exhibits strong spatial variation only in localized regions. For example, a flow field containing sharp boundary layers or localized vortices may require very high resolution in a small part of the domain, while the remaining regions vary smoothly and can be represented accurately with far fewer points. Using a uniform Cartesian grid forces the entire domain to be sampled at the highest required resolution, leading to unnecessary computational cost and memory usage.

In many applications, the geometry itself also does not naturally conform to a rectangular grid. Examples include fluid flow around airfoils, structural mechanics on curved surfaces, porous media with irregular boundaries, and many reactor physics problems involving complex geometries [5]. In such settings, the natural discretization is often given by an unstructured mesh or a point cloud rather than a regular lattice.

A common workaround is to interpolate the data from the original mesh onto a uniform Cartesian grid and then apply a standard neural operator such as the FNO or CNO. However, this interpolation step introduces additional approximation error and may discard important geometric information contained in the original discretization. In particular, regions requiring high local resolution may be smoothed or under-resolved when forced onto a regular grid, reducing both accuracy and efficiency.

These limitations motivate the development of mesh-free neural operators, which aim to learn directly from irregular point clouds or unstructured meshes without requiring interpolation onto a Cartesian grid. Such models seek to preserve the geometric information present in the original discretization while retaining the advantages of operator learning.

2.4.2 Geometry-aware Fourier Neural Operator – geo-FNO

Architecture

The geometry-aware Fourier Neural Operator (Geo-FNO), introduced by Li et al. [16], extends the original FNO to problems defined on irregular domains such as point clouds and unstructured meshes. While the standard FNO assumes that the input field is represented

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on a uniform Cartesian grid, Geo-FNO is designed to work directly with non-uniformly sampled input data.

Conceptually, Geo-FNO can be viewed as a generalization of the original FNO. The core Fourier Neural Operator remains unchanged: the model performs spectral convolution by transforming the input into Fourier space, applying learned complex weights to the retained modes, and transforming back to the spatial domain. The main modification is the introduction of a learned coordinate transformation placed before the FNO, together with non-uniform Fourier transforms at the input and output stages.

It is useful to distinguish between three different coordinate systems used in the model. The first is the input grid, corresponding to the original irregular point cloud where the data is given. The second is the computational grid, obtained after applying the learned coordinate transformation ϕ^{-1} . This grid is still non-uniform, but is mapped into a representation that is more suitable for Fourier learning. The notation ϕ^{-1} is used to reflect that the learned transformation maps from the physical (irregular) domain to a more structured computational representation. In classical settings, a mapping ϕ is typically defined from a reference or structured domain to a physical domain, and the operation required here is therefore its inverse. In practice, only ϕ^{-1} is learned and used; the forward map ϕ is never explicitly constructed. The third coordinate system is the internal grid, which is the uniform Cartesian grid used within the FNO after the first spectral layer.

The learned coordinate transformation is denoted by ϕ^{-1} , implemented in the original work as the network IPHL. Given input coordinates x , and optionally a vector of geometry parameters g , the mapping is defined as

$$\phi^{-1}(x, g) = x + f(x, g), \quad (2.21)$$

where $f(x, g)$ is represented by a lightweight multilayer perceptron and initialized close to zero so that the mapping begins near the identity transformation. The coordinate transform is trained jointly with the FNO and learns a mapping that minimizes the prediction error while adapting to the geometry of the problem.

Importantly, this is a continuous-to-continuous mapping rather than interpolation onto a discrete grid. The transformed points remain irregularly distributed, and the purpose of the mapping is not to force the domain onto a regular mesh, but to provide a coordinate system in which Fourier-based operator learning becomes more effective.

After this transformation, the input field is processed using a non-uniform discrete Fourier transform rather than the standard FFT used in the original FNO. This is necessary because the computational grid is still non-uniform. After the first spectral layer, when the inverse transform is applied, the representation is reconstructed on the uniform internal grid. The remaining Fourier layers then operate exactly as in the standard FNO using efficient FFT-based convolutions.

At the output stage, the model allows predictions to be made at arbitrary spatial locations rather than only on a fixed output mesh. During inference, a set of query points is provided corresponding to the locations where the solution is desired. These points are passed through the same learned coordinate transformation ϕ^{-1} , mapping them into the computational coordinate system.

The final layer of the FNO produces a spectral representation of the solution in terms of Fourier coefficients on the internal grid. Instead of reconstructing the solution only on that grid, the inverse transform can be evaluated directly at the transformed query points. In this way, the model can predict the solution at locations that were not explicitly present during training, including new sensor positions, boundary points, or entirely different point clouds.

This means that Geo-FNO satisfies two of the three neural operator conditions discussed in Section 2.1.2: it can accept inputs defined on irregular point sets and can be evaluated at arbitrary output positions.

Strengths

The primary strength of Geo-FNO is that it extends Fourier-based operator learning to irregular domains without requiring the full problem to be pre-interpolated onto a Cartesian grid before learning begins. This allows the model to work directly on point clouds and unstructured meshes while preserving geometric information that may be lost during interpolation to a regular lattice.

Because the internal operator remains an FNO, Geo-FNO retains many of the advantages of spectral operator learning. It still benefits from global receptive fields, efficient FFT-based convolutions in the internal layers, and strong performance on operators dominated by large-scale smooth structure. Compared to fully mesh-based or graph-based approaches, this often provides significantly better computational efficiency.

The learned coordinate transform provides an important representational advantage. Rather than relying on a fixed hand-designed mapping, the model learns a problem-dependent coordinate system jointly with the operator itself. This allows the geometry to be adapted so that the retained low Fourier modes capture more of the physically relevant information, improving approximation quality without requiring a large increase in retained modes.

Another important strength is output flexibility. Since predictions are reconstructed from Fourier coefficients rather than tied to a fixed output mesh, the model can be evaluated at arbitrary query locations. This is particularly useful for scientific computing problems where predictions may be required on new sensor locations, boundary points, or adaptive meshes not seen during training.

Weaknesses

The main limitation of the Geo-FNO is that it inherits many of the weaknesses of the original FNO. Spectral truncation remains central to the architecture, meaning that high-frequency information is still discarded and the model retains a strong low-frequency bias. For problems dominated by sharp gradients, discontinuities, or highly localized effects, this can still lead to poor approximation quality.

A second challenge arises from the use of non-uniform Fourier transforms. On a uniform Cartesian grid, the Fourier basis is orthogonal under the discrete inner product, meaning that each Fourier coefficient corresponds cleanly to a single frequency mode. Truncating high modes therefore has a clear interpretation: specific frequencies are removed from the representation.

On an irregular point cloud, this orthogonality is lost [10]. The Fourier basis functions are no longer orthogonal under sampling, and the resulting coefficients become mixtures of multiple true modes rather than clean representations of individual frequencies. As a result, truncating a single Fourier coefficient no longer corresponds to removing one specific mode, but instead removes part of several mixed components. Likewise, applying learned weights to a given coefficient affects a mixture of spectral information rather than a single interpretable frequency. This makes spectral truncation significantly less stable and reduces both computational efficiency and spectral interpretability compared to the standard FNO.

A further limitation is that the success of Geo-FNO depends heavily on the learned coordinate transform. If the mapping learned by ϕ^{-1} fails to produce a representation where the retained low Fourier modes are sufficiently informative, the model may still suffer from poor truncation behavior. Since this mapping is learned jointly with the FNO, training can become sensitive to initialization and optimization choices.

Although Geo-FNO avoids explicit pre-interpolation onto a Cartesian grid, it does not completely remove representation dependence. The quality of the learned operator still depends strongly on the spatial distribution of the input points. Poor geometric coverage, highly clustered sampling, or insufficient resolution near important local features may all produce unstable spectral representations, even after coordinate transformation. As a result, model performance depends not only on how many points are available, but critically on where those points are located relative to the underlying physics.

2.5 Discretization Invariance on Mesh-Free Grids

So far, discretization invariance has primarily been discussed in the context of functions represented on uniform Cartesian grids. In that setting, the concept is relatively straightforward: increasing the resolution corresponds to decreasing the grid spacing while preserving the same underlying domain. A model can therefore be tested by training on one resolution and evaluating on another, allowing discretization invariance to be interpreted as robustness to resolution transfer.

For non-uniform meshes and point clouds, this notion does not transfer directly. There is no single grid spacing and therefore no unique definition of resolution. Increasing the number of points alone does not necessarily improve the representation of the underlying function; what matters is where those points are placed relative to the important features of the solution. A point cloud with dense sampling near sharp gradients or boundaries may resolve the problem better than a uniformly distributed cloud with many more total points.

As a result, discretization invariance in the mesh-free setting is better interpreted as robustness to changes in sampling geometry rather than transfer across resolutions. The relevant question is not whether a model trained at one resolution generalizes to another, but whether it produces consistent and accurate predictions across different point distributions representing the same underlying problem.

For this reason, no direct resolution-transfer experiments were conducted for the mesh-free datasets considered in this thesis. Unlike the structured-grid setting used for the FNO and CNO, there is no well-posed notion of systematically increasing or decreasing resolu-

tion while preserving a comparable discretization. Varying only the number of points would not provide a meaningful measure of discretization invariance without also controlling for point placement.

Instead, the evaluation of Geo-FNO focuses on whether the model can produce physically meaningful predictions directly on irregular point clouds and whether learning on the original mesh improves performance compared to interpolation onto a uniform grid. Successful prediction in this setting indicates that the model is able to learn a stable approximation of the underlying operator despite the absence of a fixed uniform discretization.

Chapter 3

Method

This chapter describes the datasets and training procedures used throughout the thesis. It also clarifies how models with different output-resolution behavior are compared in the discretization transfer experiments.

3.1 Datasets

The experiments in this thesis use both uniform-grid and non-uniform datasets. For each dataset, the goal of this section is to describe the underlying physical problem, the mapping learned by the model, the data source, and the aspects of the dataset that are relevant for interpreting the results. Dataset-specific implementation details are included only when they affect the conclusions drawn later in the thesis. In particular, the full generation procedure for the Navier-Stokes dataset generated in this work is deferred to Appendix B.

3.1.1 Uniform-Grid Datasets

The discretization invariance experiments on regular grids use two datasets: a Navier-Stokes dataset generated in this work and a Darcy flow dataset obtained from prior work.

Navier-Stokes

For the regular-grid experiments, we use a two-dimensional Navier-Stokes dataset on the unit square. The Navier-Stokes equations are among the fundamental equations of fluid dynamics and arise in many areas of physics and engineering, including atmospheric and oceanic flows, aerodynamics, and industrial transport processes. The dataset is formulated and solved in vorticity form, with governing equation:

$$\partial_t \omega + u \cdot \nabla \omega = \nu \Delta \omega + f, \quad (3.1)$$

where ω denotes the vorticity field, u the incompressible velocity field, ν the viscosity, and f an external forcing term. The learning task considered in this thesis is not a pressure or velocity prediction problem, but an operator mapping from an initial vorticity field to a later vorticity field. The input to the model is the initial vorticity field $\omega(\cdot, 0)$, and the

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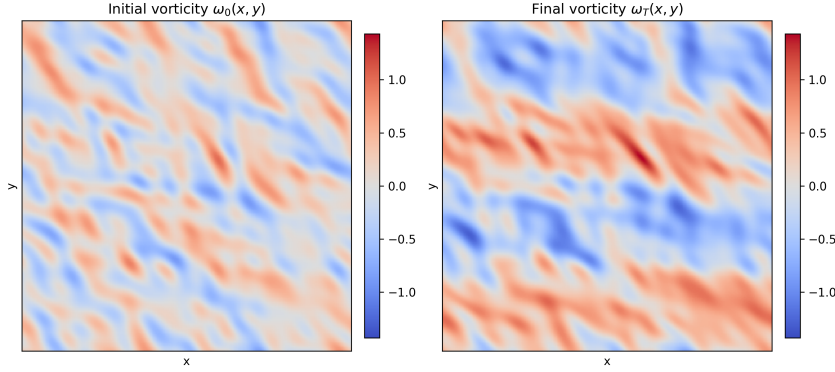


Figure 3.1: Initial and final vorticity fields for a sample from the Navier-Stokes dataset.

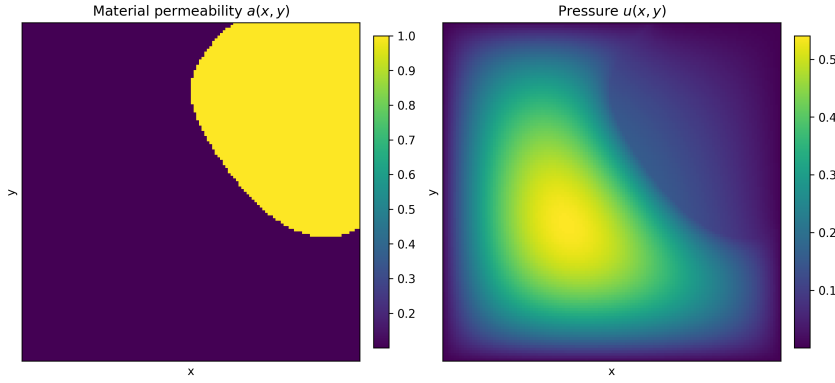


Figure 3.2: Example of an input diffusion coefficient field $a(x, y)$ and the resulting pressure field $u(x, y)$ with Dirichlet boundary conditions.

output is the vorticity field at a later time $\omega(\cdot, T)$. An example input-output pair is shown in Figure 3.1.

The dataset consists of 1000 samples generated at a resolution of 256×256 . Between samples, the viscosity and forcing remain fixed, while the initial vorticity field is varied. Further details on the data-generation procedure are provided in Appendix B.

Darcy flow

Also used for the uniform grid experiments is the Darcy flow dataset introduced by Li et al. [15]. In this benchmark, the task is to learn the solution operator of a steady-state two-dimensional Darcy flow problem on the unit square, governed by the following PDE:

$$-\nabla \cdot (a(x) \nabla u(x)) = f(x), \quad x \in (0, 1)^2, \quad (3.2)$$

$$u(x) = 0, \quad x \in \partial(0, 1)^2, \quad (3.3)$$

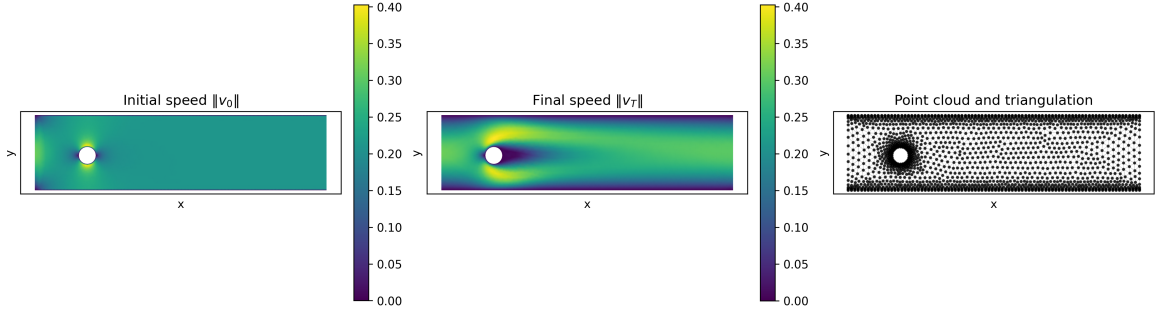


Figure 3.3: Example initial and final states for the incompressible flow past a cylinder dataset, shown on the irregular geometry together with the underlying point cloud. While the model is tasked with predicting the v_x and v_y components of the velocity field, these plots display the speed $\|\mathbf{v}\|$.

The input is a spatially varying diffusion coefficient field $a(x)$, and the output is the corresponding solution field $u(x)$ under a prescribed forcing term and Dirichlet boundary conditions. An example input-output mapping can be seen in figure 3.2. Between samples, $f(x)$ is kept constant while the $a(x)$ is varied. This PDE has many applications within physics and engineering, including drainage and irrigation in agricultural soils and fluid migration around nuclear waste repositories [17]. For this thesis, the Darcy flow dataset serves as a regular-grid benchmark for evaluating the performance of FNO and CNO across discretizations. This dataset was selected because it provides a comparatively smooth benchmark, complementing the more irregular and turbulent behavior of the Navier-Stokes dataset discussed in Section 3.1.1. Using both datasets makes it possible to evaluate model performance under qualitatively different solution regimes. The dataset contains 10000 samples at a resolution of 128×128 .

3.1.2 Non-Uniform Datasets

The experiments on irregular domains primarily use two datasets: an incompressible Euler dataset modeling flow past a cylinder and a nuclear engineering dataset consisting of temperature profiles for a single HTGR column.

Incompressible Navier-Stokes flow past a cylinder

A second non-uniform benchmark is an incompressible Navier-Stokes dataset describing fluid flow past a cylinder. This dataset was used in prior work on operator learning for PDEs and is included here because it provides a fluid-dynamics test case on a geometry that is not naturally aligned with a regular grid [13, 21].

The dataset describes two-dimensional incompressible flow with velocity $\mathbf{v}$ and pressure p , governed by:

$$\nabla \cdot \mathbf{v} = 0, \quad (3.4)$$

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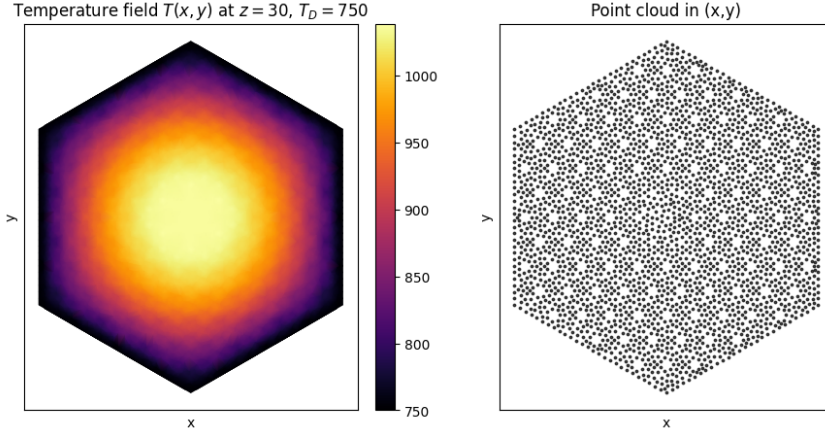


Figure 3.4: Example temperature field from the HTGR column dataset for a fixed boundary temperature and selected axial layer, shown together with the irregular spatial mesh used for sampling.

$$\rho_0 (\partial_t \mathbf{v} + \mathbf{v} \cdot \nabla \mathbf{v}) + \nabla p = \mu \nabla^2 \mathbf{v}, \quad (3.5)$$

where ρ_0 is the constant density and μ the viscosity. The domain consists of a channel containing a cylindrical obstacle, with the cylinder position and diameter varying between samples.

The prediction task is to learn a mapping from the velocity field at the initial time to the velocity field at the final time, while accounting for the fact that changes in cylinder size and location lead to different corresponding flow patterns. In the experiments of this thesis, the model is trained to predict only the horizontal and vertical velocity components; pressure is not included in the target.

This dataset is particularly relevant for Geo-FNO because it is defined on a non-uniform spatial discretization rather than a regular Cartesian grid. An example initial-final pair is shown in Figure 3.3. The dataset consists of 1000 samples each containing between 1700 and 2100 points, depending on the geometry of the sample. 200 of these samples were designated for testing, while the remaining 800 were used for training.

HTGR column temperature dataset

The final non-uniform dataset consists of steady-state temperature profiles for a single column of a prismatic core high temperature gas-cooled reactor (HTGR). This dataset is included because it provides a first application-oriented test case for Geo-FNO in a nuclear engineering setting. Unlike the public benchmark datasets considered earlier, the data was produced as part of ongoing work by a colleague within the RPNM group.

Physically, the dataset describes the temperature distribution within a three-dimensional reactor column. For each case, a constant boundary temperature is imposed and a coupled neutronics-thermohydraulics calculation is performed until a steady state is reached. The transient evolution toward this steady state is not included in the dataset; only the final

steady-state temperature fields are used. The use of a fixed boundary temperature is a simplifying assumption, since in a realistic reactor system the boundary temperature would itself change as the system evolves. This simplification was made to produce a controlled dataset within the time constraints of the project.

The spatial sampling locations lie on an irregular mesh that is shared across all samples. Along the axial direction, the column is quantized into 17 distinct z -levels. The imposed boundary temperature ranges from 300°C to 1000°C in steps of 1°C , yielding 701 distinct thermal conditions. Since the prediction task is defined on individual horizontal slices rather than on the full three-dimensional column, this results in a total of $17 \times 701 = 11917$ two-dimensional samples.

In the prediction task considered in this thesis, the model receives the boundary temperature together with a specified z -level and predicts the temperature at all sampled (x, y) locations in the corresponding horizontal slice. The task can therefore be interpreted as learning a mapping from the boundary condition and axial position to the two-dimensional temperature profile within that layer. An example of such a slice, together with the underlying irregular mesh, is shown in Figure 3.4.

3.2 Training Setup

All models are trained in a supervised setting on paired input-output samples from the datasets described above. For the uniform experiments, the data is split into a training set and a held-out evaluation set. The training set is used to optimize the model parameters, while the held-out set is used to monitor performance during training and to report final model performance. For the non-uniform experiments, a validation set was also included, with all the listed losses being from this set.

The hyperparameters differ between the uniform-grid and non-uniform-grid experiments, as well as between datasets. To keep the main text focused on the experimental design rather than implementation details, hyperparameter settings and training details are given in Appendix D.

3.3 Uniform-Grid Resolution Transfer Experiments

The uniform-grid experiments are designed to evaluate discretization invariance by testing how well a model trained at one spatial resolution generalizes to other resolutions. The Navier-Stokes and Darcy flow datasets described in 3.1.1 are used for these experiments. For each dataset, the models are trained at a chosen training resolution and then evaluated both at the training resolution and at lower and higher resolutions. This makes it possible to compare performance at the training resolution with performance after transferring the same trained model to different spatial resolutions.

For the Navier-Stokes dataset, the models are trained at a resolution of 192×192 and evaluated on resolutions from 128×128 to 256×256 in steps of 16 grid points. For the Darcy flow dataset, the models are trained at a resolution of 96×96 and evaluated on resolutions from 64×64 to 128×128 in steps of 8 grid points.

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Four model variants are compared for each dataset. The first two are FNO models retaining either 8 or 16 Fourier modes in each spatial direction, denoted FNO_k8 and FNO_k16. The other two are CNO variants. The first is the regular CNO architecture as outlined in section 2.3, referred to as CNO_standard. The second is a modified CNO in which the upsample-nonlinearity-downsample (UND) operation is replaced by a simple pointwise nonlinearity, referred to as CNO_no_UND. This operation is one of the architectural changes that distinguishes the CNO from a U-Net-like architecture and is intended to reduce aliasing effects introduced by nonlinearities. Comparing the standard and modified CNO therefore makes it possible to assess the effect of this component in the resolution-transfer setting.

FNO and CNO handle changes in resolution differently. FNO produces predictions at the same resolution as the input it is given. When an FNO is evaluated at a resolution $N \times N$, both the input and target field are resampled to $N \times N$, after which the FNO prediction can be compared directly to the target field on that grid. CNO, however, produces predictions on its internal resolution. Therefore, when CNO is evaluated on an input resolution different from its internal resolution, the CNO prediction is resampled onto the input resolution before the error is computed. The target field is resampled to the same resolution, so that both FNO and CNO are evaluated against target fields on the same spatial grid. All resampling in these experiments is performed with bicubic interpolation.

In addition to reporting the loss at each tested resolution, a scalar measure is used to summarize how strongly the loss changes away from the training resolution. For a set of tested resolutions R with loss L_N at resolution N^1 and with training resolution N_{train} , this score is defined as:

$$D = \frac{1}{|R|} \sum_{N \in R} |\log(L_N) - \log(L_{N_{\text{train}}})|. \quad (3.6)$$

Lower values of D indicate that the loss remains closer to the loss at the training resolution across the evaluated resolutions. This score is only a measure of consistency across resolutions, not of absolute model quality. The loss values themselves remain the primary measure of predictive performance, while D is used only as a quantitative measure of discretization robustness.

3.4 Non-Uniform-Grid Geo-FNO Evaluations

The non-uniform-grid evaluations use Geo-FNO on the cylinder flow and HTGR column temperature datasets described in Section 3.1.2. These evaluations differ from the uniform-grid resolution transfer experiments in Section 3.3. Since the spatial samples are given as point clouds or irregular meshes rather than Cartesian grids, there is no single resolution parameter equivalent to the grid size N used in the uniform-grid experiments. The goal is therefore not to measure resolution transfer, but to evaluate whether Geo-FNO can learn mappings on irregular spatial data.

¹ All evaluated domains are square. The resolution is therefore reported by its one-dimensional grid size N , while the full discretized domain contains N^2 grid points.

Feature engineering was used in the non-uniform experiments to augment the raw model inputs with additional variables that make relevant structure in the data more explicit. In general, feature engineering refers to the transformation or augmentation of raw input variables into representations that are more informative for a machine learning model, and prior work has shown that such transformations can improve predictive performance [11].

For the cylinder flow dataset, the model did not produce meaningful predictions when it was given only the velocity field on the irregular mesh. The coordinate transform MLP ϕ^{-1} was given the x and y coordinates of the cylinder’s center and its radius. In addition to the horizontal and vertical velocity components, each mesh point was also assigned a one-hot encoding of its node type. The node-type categories distinguish the upper, lower and cylinder boundary nodes, left boundary nodes, right boundary nodes, and interior nodes. This made the location of the obstacle more explicit in the pointwise input representation, allowing the model to identify which mesh points lie on or near the cylinder.

For the HTGR column temperature dataset, the raw input consists only of the boundary temperature T and the axial level z . Since this is a low-dimensional representation of the input conditions, several alternative representations of T and z were tested. These input representations and their effect on model performance are discussed in Appendix C. The best performing representation was a trigonometric encoding of z and T and was used for the results reported in the main text.

For the HTGR column temperature dataset, a separate train-validation-test split is used. The split is constructed over individual two-dimensional slice samples, where each sample corresponds to one combination of boundary temperature and axial level. Unlike a random split, the validation and test samples are selected using fixed subsets of the (z, T) input space.

The non-uniform-grid results should therefore be interpreted as held-out-data evaluations of Geo-FNO on irregular spatial representations, rather than as a direct analogue of the uniform-grid discretization-invariance study. Implementation issues encountered during development, including the seam artifact observed in early Geo-FNO predictions, are discussed separately in Appendix E. The results in the main text are reported using the corrected implementation.

Chapter 4

Uniform Grids – Results and Discussion

4.1 Results

4.1.1 Navier-Stokes

The four models outlined in Section 3.3 were evaluated on the Navier-Stokes dataset. Figure 4.1 shows the MSE loss as a function of evaluation resolution for each model, with the training resolution indicated by the vertical dashed line. Table 4.1 reports the discretization robustness score D , which measures how much the loss changes across resolutions relative

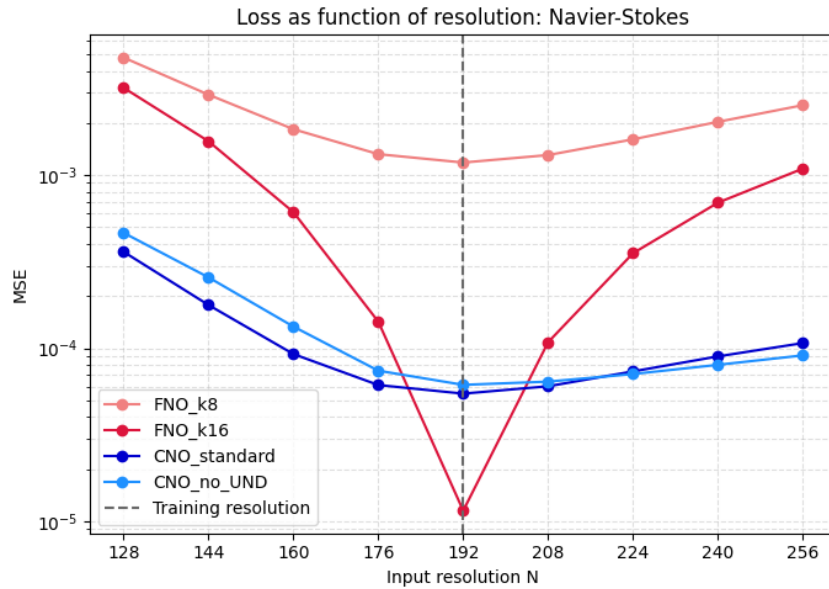


Figure 4.1: A plot of the MSE loss as a function of resolution for the four models trained on the Navier-Stokes dataset. The training resolution is indicated by the vertical dashed line.

4. UNIFORM GRIDS – RESULTS AND DISCUSSION

Table 4.1: Discretization robustness score D for the Navier-Stokes uniform-grid experiment. Lower values indicate that the loss remains more consistent across evaluation resolutions relative to the loss at the training resolution.

Model	D
FNO_k8	0.5076
FNO_k16	3.4817
CNO_standard	0.5854
CNO_no_UND	0.5847

Table 4.2: Discretization robustness score D for the Darcy flow uniform-grid experiment. Lower values indicate that the loss remains more consistent across evaluation resolutions relative to the loss at the training resolution.

Model	D
FNO_k8	0.3583
FNO_k16	0.4179
CNO_standard	0.0140
CNO_no_UND	0.0260

to the loss at the training resolution.

Among the FNO models, FNO_k16 achieves the lowest loss at the training resolution, but also has the largest value of D . Its performance therefore changes the most under resolution transfer. This contrast is especially pronounced at the lowest tested resolution, where the loss of FNO_k16 increases by more than two orders of magnitude compared with its loss at the training resolution. FNO_k8 has a higher loss at all tested resolution including the training resolution, but a substantially lower value of D , indicating a more consistent loss across the evaluated resolutions. For both FNO variants, the loss increases more rapidly when evaluating below the training resolution than when evaluating above it.

The two CNO variants show much flatter loss curves than FNO_k16 and have nearly identical values of D . Their training-resolution losses are also comparable. The standard CNO performs slightly better at the training resolution, at lower resolutions, and at $N = 208$. At the highest evaluated resolutions, CNO_no_UND gives slightly lower losses than CNO_standard. Overall, the Navier-Stokes results show a trade-off: FNO_k16 gives the best performance at the training resolution, while FNO_k8 and the CNO variants show more consistent behavior across resolutions.

4.1.2 Darcy flow

To test whether the trends observed on the Navier-Stokes dataset also appear on a qualitatively different PDE benchmark, the four models were evaluated on the Darcy flow dataset. The Darcy flow dataset was chosen as the solution space is smooth, making it a good point of comparison to the Navier-Stokes dataset. Figure 4.2 shows the MSE loss as a function of evaluation resolution for each model, with the training resolution indicated by the vertical

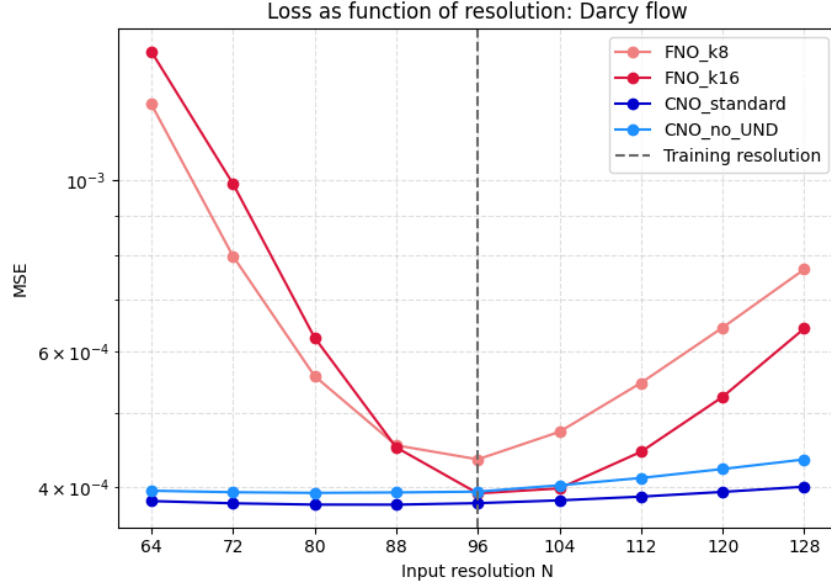


Figure 4.2: A plot of the MSE loss as a function of resolution for the four models trained on the Darcy flow dataset. The training resolution is indicated by the vertical dashed line.

dashed line. Table 4.2 reports the corresponding discretization robustness score D .

Compared with the Navier-Stokes experiment, the four models are closer in overall performance on the Darcy flow dataset. At the training resolution, CNO_standard achieves the lowest loss, followed by FNO_k16, CNO_no_UND, and FNO_k8. The two FNO variants show a similar pattern to that observed for Navier-Stokes; the loss is lowest near the training resolution and increases when the model is evaluated at other resolutions. This increase is stronger below the training resolution than above it. Notably, FNO_k8 outperforms FNO_k16 at resolutions below $N = 88$. This indicates that retaining more Fourier modes does not consistently improve performance under transfer to coarser grids.

The CNO variants show much flatter loss curves across the evaluated resolutions, with D values of order 10^{-2} . At lower resolutions, both CNO models maintain an almost constant loss, while at higher resolutions both losses increase slowly. In contrast to the Navier-Stokes experiment, CNO_standard outperforms CNO_no_UND at all evaluated resolutions. The loss of CNO_no_UND also increases more quickly than that of CNO_standard when evaluating above the training resolution. Overall, the Darcy flow results show a clearer separation between the FNO and CNO variants in terms of discretization robustness; the FNO models are more sensitive to resolution transfer, while the CNO models remain nearly constant across the tested range.

4.2 Discussion

4.2.1 FNO

The FNO variants show the weakest discretization robustness of the models tested on both datasets. This is most clearly visible for FNO_k16, which achieves the lowest loss at the training resolution on Navier–Stokes, but also the largest discretization robustness score D by a large margin. This indicates that strong in-distribution performance does not imply robustness under resolution transfer.

A central reason for this behavior is that the meaning of each retained Fourier mode depends on the spatial resolution of the input grid. During training, the FNO is only exposed to a single discretization, and therefore implicitly learns an association between a mode’s index and its physical meaning. For a fixed physical domain, a given Fourier mode index corresponds to the same nominal wavelength across resolutions. However, changing the resolution changes the highest resolvable frequency and therefore the distribution of spectral energy across the retained modes. At lower resolutions, unresolved high-frequency content may alias into the retained coefficients, while at higher resolutions that same information is represented explicitly in additional modes. As a result, the same retained mode coefficient may contain very different information depending on the discretization. When the resolution changes, this correspondence no longer holds; the same mode index no longer represents the same physical structure, and the learned spectral weights are applied to coefficients with different physical meaning than those seen during training.

This effect becomes particularly severe when evaluating below the training resolution. At lower resolutions, fewer frequencies are resolvable and some physically important modes are no longer represented correctly. High-frequency content folds into the retained modes through aliasing, and the spectral representation itself becomes corrupted. The model is therefore not only missing information, but is given misleading low-frequency content that was not present during training. In contrast, when evaluating above the training resolution, the model simply encounters additional high-frequency information that it was not trained to use. This is suboptimal, but less destructive than the irreversible information loss that occurs at coarse resolutions. This explains why the loss increases more rapidly below the training resolution than above it for both FNO variants.

The comparison between FNO_k8 and FNO_k16 highlights an important distinction between absolute accuracy and discretization robustness. Retaining more Fourier modes improves the expressive power of the model, which is why FNO_k16 achieves lower loss than FNO_k8 at the training resolution and, for the Navier–Stokes dataset, across all tested resolutions. The additional retained modes contain physically relevant information that improves prediction quality even when resolution transfer introduces mismatch.

However, these same modes are also the most sensitive to changes in discretization. Since higher-frequency coefficients are more strongly affected by aliasing, under-resolution, and changes in spectral energy distribution, relying on them makes the model less robust when evaluated away from the training grid. This is reflected in the much larger value of D for FNO_k16. FNO_k8 is therefore not better in terms of predictive accuracy, but simply less sensitive because it relies on a more aggressively truncated and therefore more stable

spectral representation.

This behavior is particularly clear on the Darcy flow dataset, where FNO_k8 outperforms FNO_k16 at sufficiently low resolutions. Darcy flow is a smooth elliptic problem with a strongly smoothing solution operator, meaning that the high-frequency content of the solution is relatively weak. The additional modes retained by FNO_k16 therefore contain limited useful physical information and are more sensitive to discretization errors, poorly resolved modes, and interpolation noise. At coarse resolutions, relying on these modes becomes detrimental, while the more aggressively truncated FNO_k8 remains more stable. At resolutions near or above training, where these modes are better resolved, the additional modes of FNO_k16 again becomes advantageous.

For the Navier–Stokes dataset, the strong performance of FNO_k16 at the training resolution is also partly artificial. As outlined in appendix B, the dataset is generated spectrally using 16 retained Fourier modes in each direction, meaning that an FNO retaining exactly 16 modes is naturally well matched to the structure of the data. This makes FNO_k16 particularly well suited to the benchmark and gives it an advantage that may not transfer to more general datasets where the true solution is not already bandlimited to the retained modes.

4.2.2 CNO

In contrast to the FNO, the CNO variants show substantially stronger discretization robustness on both datasets. Their loss curves remain much flatter across evaluation resolutions, and the corresponding values of D are significantly lower, particularly on the Darcy flow dataset where both CNO models remain nearly constant across much of the tested range.

A central reason for this behavior is the initial resampling step onto a fixed internal grid. Regardless of the input resolution, the CNO first interpolates the input field onto the same internal discretization before the main operator learning begins. This removes the resolution-dependent change in representation that causes much of the instability in the FNO. The convolutional blocks and nonlinear operations therefore always act on the same effective resolution, meaning that the model learns a more consistent operator across discretizations.

This behavior is particularly clear for the Darcy flow dataset. Darcy flow is governed by a smooth elliptic operator, and the solution space is strongly dominated by low-frequency structure. As a result, even relatively coarse evaluation grids still preserve the physically relevant information needed for prediction. Since the CNO always projects the input onto the same internal representation, evaluating at lower resolutions introduces only a small loss of information, and the model remains almost unchanged across the tested range.

At higher evaluation resolutions, the loss of the CNO increases gradually for both datasets. This is expected from the explicit bandlimited assumption built into the architecture. Because the model always resamples onto the same internal grid, frequencies beyond that effective bandlimit are removed before learning begins. Higher-resolution inputs therefore contain physically meaningful fine-scale information that the model deliberately discards. As the evaluation resolution increases, the difference between the bandlimited representation seen during training and the higher-bandwidth true field becomes larger, leading to a gradual increase in loss. Over the tested range, this increase is approximately linear.

The difference between CNO_standard and CNO_no_UND is most visible at high resolutions, particularly for the Navier–Stokes dataset where CNO_no_UND slightly outperforms the standard CNO. This behavior can be understood by considering the role of the upsample–nonlinearity–downsample (UND) block described in Section 2.3.

The purpose of the UND block is to prevent nonlinear activations from generating unresolved high-frequency content that aliases back into the retained modes. By temporarily upsampling before the activation and applying low-pass filtering during the subsequent downsampling step, the CNO preserves a more consistent bandlimited representation and improves continuum consistency.

However, the low-pass filter used during downsampling is only an approximation of the ideal sinc projection. In practice, it acts as a smoothing operation rather than a perfectly sharp spectral cutoff. While this successfully removes unresolved frequencies, it also attenuates the highest retained modes and slightly reduces access to the finest physically meaningful structures available to the model.

For Darcy flow, this has little practical consequence because the solution operator is strongly smoothing and the highest frequencies contain relatively little useful information. The standard CNO therefore performs better across all tested resolutions.

For the much more nonlinear Navier–Stokes problem, however, the highest retained modes are significantly more important. The attenuation imposed by the approximate low-pass filter effectively enforces an even stricter bandlimit on the learned representation. As evaluation resolution increases, the true solution contains progressively more physically meaningful high-frequency structure, and the loss grows according to how quickly the bandlimited training representation diverges from these higher-bandwidth fields. Because CNO_standard is effectively trained with a stronger bandlimit than CNO_no_UND, this divergence occurs faster, which explains why CNO_no_UND slightly outperforms the standard CNO at the highest tested resolutions despite being theoretically less representation-equivalent.

This does not imply that the UND block should be removed in general. Across most settings, including all Darcy flow experiments and lower-resolution Navier–Stokes evaluations, the standard CNO performs as well or better and remains the more theoretically consistent architecture. Rather, these results suggest that the practical approximation of the ideal low-pass projection can become a limiting factor for highly nonlinear problems with significant high-frequency content.

4.2.3 Improving the FNO

To identify which parts of the CNO architecture are primarily responsible for its stronger discretization robustness, an ablation study was performed on the FNO. The two features hypothesized to be most important were resampling onto a fixed internal grid and the UND block. Rather than introducing both modifications at once, each was tested individually as well as in combination.

Four models were therefore compared: the standard FNO, FNO_UND, which adds only the UND block, FNO_resample, which adds only the fixed-grid resampling step, and

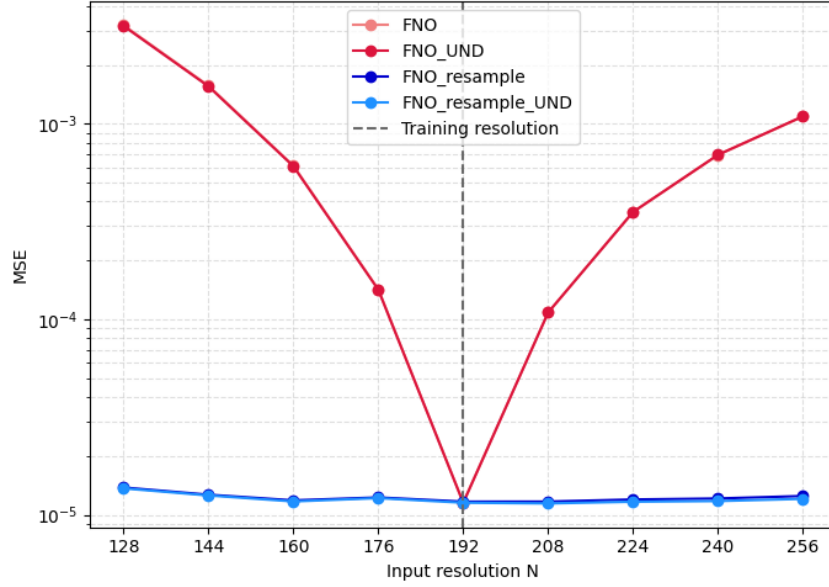


Figure 4.3: A plot of the MSE loss for four FNOs trained on the Navier-Stokes dataset. FNO is the standard FNO; the other models have been modified to include resampling onto a fixed internal grid and/or use the UND block used by the CNO. The training resolution is indicated by the vertical dashed line.

Table 4.3: Discretization robustness score D for the FNOs modified to improve discretization robustness on the Navier-Stokes dataset.

Model	D
FNO	3.4817
FNO_UND	3.4800
FNO_resample	0.0490
FNO_resample_UND	0.0456

FNO_resample_UND, which includes both modifications. All models retained 16 Fourier modes in each direction and were trained and evaluated on the Navier-Stokes dataset.

The results are shown in Figure 4.3, with the corresponding discretization robustness scores reported in Table 4.3. One observation is that the UND block alone produces only a negligible improvement. The values of D for FNO and FNO_UND are nearly identical, indicating that while aliasing from nonlinear activations contributes to discretization mismatch, it is not the dominant failure mechanism responsible for the poor robustness of the standard FNO.

In contrast, introducing resampling onto a fixed internal grid produces a dramatic improvement. Both FNO_resample and FNO_resample_UND reduce the discretization robustness score by nearly two orders of magnitude compared to the standard FNO. This strongly supports the interpretation developed in the previous sections: the primary source of insta-

bility in the FNO is not the nonlinearity itself, but the dependence of the retained Fourier coefficients on the input discretization.

Without resampling, the model learns on a fixed set of retained Fourier modes whose coefficients change meaning when the input resolution changes. Although the nominal wavelength associated with a mode index remains the same, the highest resolvable frequency changes with resolution, which alters the spectral energy distribution and introduces aliasing errors at coarse resolutions. As a result, the same retained coefficient may contain very different information depending on the discretization, causing the learned spectral weights to become inconsistent under resolution transfer.

Resampling resolves this problem by enforcing a fixed internal representation before the Fourier layers are applied. Regardless of the input resolution, the model always operates on the same effective discretization and therefore sees the same bandlimit and the same spectral structure during both training and inference. This restores the consistency between retained mode coefficients and their physical meaning, making the learned operator substantially more robust across resolutions.

The remaining small improvement from adding the UND block on top of resampling is consistent with the role of nonlinear aliasing described earlier. Once the dominant representation mismatch is removed by resampling, controlling the frequencies generated by nonlinear activations provides a small additional benefit, but does not fundamentally change the behavior of the model.

It is important to remember that the Navier–Stokes benchmark used here favors the FNO somewhat artificially. The initial conditions are generated spectrally using 16 retained Fourier modes in each direction, meaning that an FNO retaining exactly 16 modes is naturally well matched to the structure of the data. As a result, all tested resolutions are sufficiently fine to capture the full signal, and the resampled FNO does not show the gradual high-resolution loss increase observed for the CNO. For more general datasets where the true solution is not already bandlimited to the retained modes, a gradual increase in loss at higher resolutions would still be expected, governed by how quickly the fixed training representation diverges from higher-bandwidth fields as was seen for the CNO.

Overall, these results show that the dominant source of poor discretization robustness in the FNO is the resolution dependence of its spectral representation. By introducing the fixed internal resampling strategy used in the CNO, the FNO becomes dramatically more stable under resolution transfer, while the UND block provides a smaller secondary improvement.

Chapter 5

Non-Uniform Meshes – Results and Discussion

5.1 Results

5.1.1 Cylinder flow

The Geo-FNO was first evaluated on the cylinder flow dataset. Figure 5.1 shows a representative prediction on a held-out test sample.

The prediction captures the main structure of the final velocity field, including the global flow pattern around the cylinder and the downstream wake. The difference plot shows that the error is not distributed uniformly across the domain. Larger errors occur primarily in regions where the velocity field changes more sharply, especially near the wake and around the obstacle. Despite these local errors, the qualitative agreement between the prediction and the ground truth indicates that Geo-FNO can learn the velocity-field mapping directly on the irregular point-cloud representation.

5.1.2 HTGR column temperature dataset

Geo-FNO was evaluated on the HTGR column temperature dataset described in Section 3.1.2. Figure 5.2 shows a representative prediction on a held-out sample.

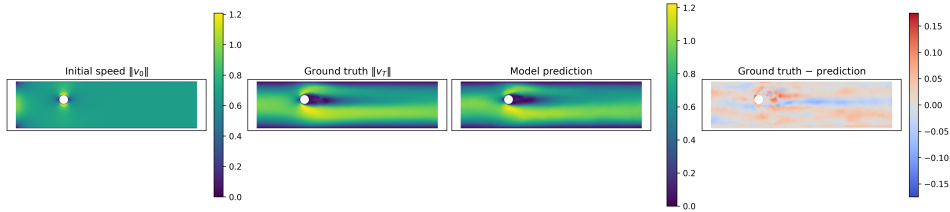


Figure 5.1: Representative Geo-FNO prediction on the cylinder flow dataset. The panels show the initial velocity field, the ground-truth final velocity field, the model prediction, and the pointwise difference between prediction and ground truth.

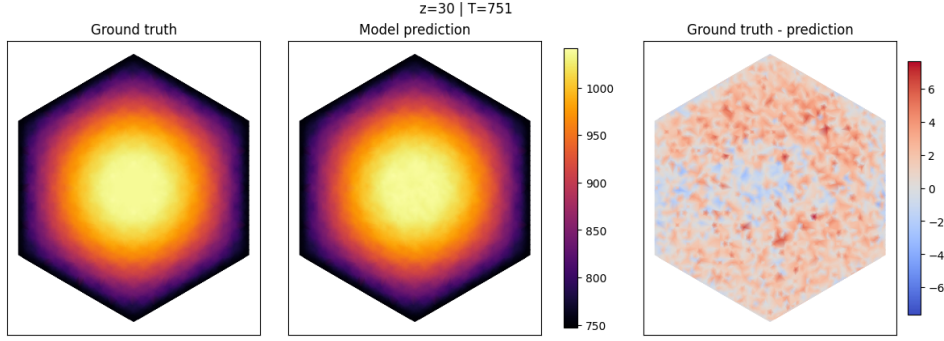


Figure 5.2: Representative Geo-FNO prediction on the HTGR column temperature dataset. The panels show the ground-truth temperature field, the model prediction, and the difference between prediction and ground truth on the irregular spatial mesh.

The predicted temperature field closely matches the ground truth. The dominant spatial temperature pattern is reproduced, and the difference plot shows small local deviations relative to the overall temperature scale.

5.2 Discussion

5.2.1 Cylinder Flow

The cylinder flow dataset provides a strong benchmark for mesh-free operator learning because the model must solve two difficulties simultaneously: learning nontrivial fluid dynamics and doing so directly on an irregular spatial representation. Unlike the uniform-grid experiments in the previous chapter, the geometry itself varies between samples through changes in the cylinder diameter, and the solution must be predicted on a non-uniform point cloud rather than on a fixed Cartesian grid. The task is therefore not only to learn the evolution of the velocity field, but also to remain consistent across changing geometries and irregular sampling patterns.

Since the inlet speed and viscosity are fixed across the dataset, the Reynolds number is directly proportional to the cylinder diameter,

$$Re = \frac{UD}{\nu}, \quad (5.1)$$

where U is the inlet speed, D is the cylinder diameter, and ν is the kinematic viscosity [24]. Larger cylinders therefore correspond to higher Reynolds numbers and stronger wake instability, making the prediction task progressively more difficult as the obstacle size increases.

Figure 5.1 shows that Geo-FNO captures the dominant structure of the flow well, including the global velocity field around the obstacle and the downstream wake. The largest prediction errors are concentrated in the wake region behind the cylinder rather than near the obstacle itself. This is physically reasonable, since the downstream wake is highly sensitive to small perturbations and contains the most dynamically complex part of the solution.

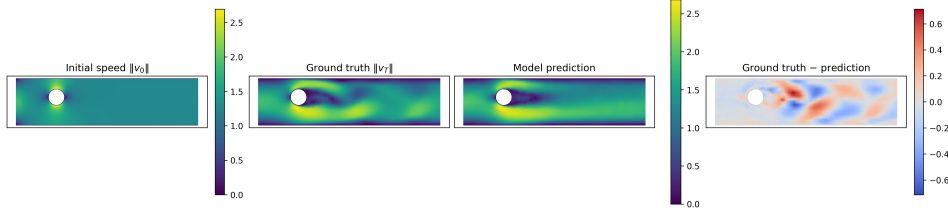


Figure 5.3: Representative Geo-FNO prediction on the cylinder flow dataset for a sample consistent with periodic vortex shedding in the wake region. The panels show the initial velocity field, the ground-truth final velocity field, the model prediction, and the pointwise difference between prediction and ground truth. Unlike the near-steady case, the downstream wake remains time-periodic and oscillates around a mean position. The model accurately captures the overall flow structure but predicts a smoothed mean wake rather than the exact instantaneous phase of the oscillatory tail, leading to larger localized errors downstream of the cylinder.

For some samples, the dominant prediction error occurs in the wake region downstream of the cylinder, where the flow does not appear to converge to a simple steady wake structure. Instead, the ground truth shows a more oscillatory downstream pattern, while the model tends to predict a smoother mean wake. This behavior is illustrated in Figure 5.3. This is consistent with an unsteady wake regime, which may arise from several causes.

One possibility is that some samples have not fully relaxed to a steady solution by the final simulation time, so the model is asked to predict a transient state rather than a converged final configuration. Another possibility is that, for larger cylinders corresponding to higher Reynolds numbers, the wake enters an oscillatory regime associated with periodic vortex shedding rather than converging to a static steady state [23]. In the dataset, these more oscillatory wake structures are observed primarily for the larger cylinders, which makes vortex shedding the most likely explanation.

Regardless of the cause of the oscillatory behavior, the difficulty remains the same: the model performs one-shot prediction of the final state directly from the initial condition, rather than explicitly evolving the flow through time. Predicting the exact instantaneous state of an unsteady wake is therefore significantly more difficult than predicting its mean structure, since small differences in phase can produce large pointwise differences downstream while leaving the overall flow pattern largely unchanged.

As a result, the model often captures the correct average wake location while smoothing out the instantaneous downstream variations. This leads to relatively high loss despite strong qualitative agreement with the overall flow field, particularly in the wake region where the solution is most sensitive to transient effects.

These results suggest that a significant portion of the remaining error is associated with predicting the exact instantaneous phase of an unsteady wake rather than with failure to capture the dominant flow physics. Geo-FNO successfully learns the global flow structure and the mean wake behavior, but pointwise error metrics such as MSE penalize phase shifts in the downstream oscillatory region heavily, even when the overall physical prediction is

qualitatively correct. This makes one-shot surrogate prediction of the final state fundamentally more difficult than autoregressive rollout approaches that explicitly evolve the flow through time.

5.2.2 HTGR Column Temperature Dataset

The strong performance of Geo-FNO on the HTGR column temperature dataset should be interpreted in the context of the structure of the task. Compared to the cylinder flow benchmark, this problem is both smoother and lower-dimensional. The output field is a scalar temperature distribution rather than a vector-valued velocity field, and the dominant physical behavior is governed by diffusion and heat transport rather than strongly nonlinear wake dynamics.

In addition, the spatial mesh is shared across all samples. Unlike the cylinder flow dataset, where both the geometry and the point cloud vary between samples, the Geo-FNO is always evaluated on the same underlying irregular mesh. The model therefore does not need to adapt to changing geometries or different point distributions, and the challenge is focused primarily on learning the temperature field itself rather than learning across varying spatial representations.

The input space is also significantly simpler. As described in Section 3.1.2, the prediction depends only on the boundary temperature and the axial position of the slice¹. This makes the task closer to learning a smooth parametric family of temperature fields than to solving a fully general operator learning problem with high-dimensional functional inputs.

Figure 5.2 shows that Geo-FNO reproduces the dominant temperature structure very accurately. The prediction captures both the global temperature gradient and the local spatial variation across the irregular mesh, while the difference plot shows only small local deviations relative to the overall temperature scale.

These results show that Geo-FNO can be applied successfully to reactor-relevant temperature fields defined on non-uniform meshes without requiring interpolation onto a uniform Cartesian grid. This is useful in practice, since reactor geometries often involve curved boundaries and spatial layouts that are not naturally represented on regular grids.

At the same time, the relative simplicity of the HTGR task should be kept in mind. Because the geometry is fixed, the field is smooth, and the input space is low-dimensional, this experiment is best interpreted as a proof of applicability rather than a demonstration of performance on more complex reactor simulations. It shows that Geo-FNO can operate effectively on irregular reactor meshes, but does not by itself establish performance on fully general multiphysics reactor problems.

5.2.3 Interpretation of Discretization Invariance on Mesh-Free Grids

As discussed in Section 2.5, discretization invariance on non-uniform meshes cannot be interpreted through resolution transfer in the same way as on uniform Cartesian grids. Since there is no single notion of grid spacing for a point cloud, increasing the number of points

¹As discussed in Appendix C, a trigonometric representation of z and T produced the lowest loss and was therefore used for the results reported here.

alone does not provide a meaningful measure of resolution unless the spatial distribution of those points is also controlled. For this reason, no direct analogue of the resolution-transfer experiments performed for the FNO and CNO was conducted for Geo-FNO.

Instead, the mesh-free experiments evaluate whether the model can produce stable and physically meaningful predictions directly on irregular spatial representations and across changing geometries. In the cylinder flow dataset, both the geometry and the point cloud vary between samples, making generalization significantly more demanding than on a fixed mesh. Successful prediction in this setting therefore provides evidence that the model is learning an operator rather than memorizing a particular discretization. In the HTGR dataset, the mesh is fixed, but the results demonstrate that Geo-FNO can still be applied effectively to reactor-relevant fields defined on non-uniform domains without requiring interpolation onto a uniform Cartesian grid.

This also connects to the characterization of neural operators introduced in Section 2.1.2. Geo-FNO naturally satisfies discretization flexibility and pointwise evaluation more directly than standard uniform-grid architectures, since it can accept input data defined on irregular point clouds and produce predictions at arbitrary query locations. Continuum consistency, however, is more difficult to verify experimentally in the mesh-free setting because there is no canonical notion of systematic mesh refinement comparable to uniform-grid resolution transfer.

The results of this chapter should therefore not be interpreted as a direct demonstration of discretization invariance in the same sense as the uniform-grid experiments. Rather, they show that Geo-FNO is robust to irregular sampling and geometric variation while maintaining meaningful predictive performance. In the mesh-free setting, this robustness is the practical analogue of discretization invariance: the ability to learn a stable approximation of the underlying continuous operator despite the absence of a fixed regular discretization.

Chapter 6

Conclusions and Future Work

6.1 Conclusions

The goal of this thesis was to investigate to what extent Fourier- and convolution-based neural operators can provide discretization-robust surrogate models for PDE-based data on both uniform grids and non-uniform point clouds. Particular focus was placed on discretization invariance: whether a model learns the underlying continuous operator or becomes tied to the specific discretization used during training.

For uniform Cartesian grids, the results show a clear difference between the FNO and the CNO. While the FNO achieves strong accuracy at the training resolution, its performance degrades significantly under resolution transfer, especially when evaluated on coarser grids. This behavior is caused primarily by the dependence of retained Fourier coefficients on the training discretization. As the evaluation resolution changes, the physical meaning and energy distribution of the retained modes change, causing the learned correspondence between Fourier coefficients and physical structure to break down.

The comparison between FNO_k8 and FNO_k16 further highlights this trade-off. Retaining more modes improves expressivity and lowers the in-distribution loss, but also increases sensitivity to discretization mismatch. The stronger performance of FNO_k16 at all tested resolutions shows that additional spectral information is useful, but its much larger increase in loss away from the training resolution demonstrates that this improved accuracy comes at the cost of robustness.

In contrast, the CNO exhibits substantially stronger discretization robustness across both Navier–Stokes and Darcy flow. Its use of fixed internal resampling provides a stable representation across resolutions, while the multi-scale encoder–decoder structure allows the model to capture both local and global behavior without relying on resolution-dependent Fourier modes. The results show that this architectural choice is more important for robustness than the specific use of spectral representations.

The role of the upsample–nonlinearity–downsample (UND) block was found to be more subtle. The UND block improves mathematical consistency by preventing nonlinear aliasing from contaminating the retained modes, but it is not the dominant factor controlling discretization robustness. In the modified FNO experiments, introducing UND alone pro-

duced only a negligible improvement, while resampling onto a fixed internal grid reduced discretization sensitivity by nearly two orders of magnitude. This shows that representation consistency is the primary mechanism governing robustness, while aliasing control is a secondary but still theoretically important refinement.

For non-uniform meshes, the results show that the Geo-FNO successfully extends operator learning to irregular spatial domains. On both the cylinder flow dataset and the HTGR temperature dataset, the model produced physically meaningful predictions directly on point clouds and non-uniform meshes without requiring interpolation onto a uniform Cartesian grid. This demonstrates that Fourier-based operator learning can remain effective even when the assumptions of the standard FNO no longer hold.

At the same time, the thesis shows that discretization invariance must be interpreted differently in the mesh-free setting. On uniform grids, robustness can be evaluated through resolution transfer by changing grid spacing while preserving the same domain. For point clouds, there is no single notion of resolution; the quality of the representation depends not only on the number of points, but also on where those points are placed relative to important physical features. As a result, mesh-free discretization invariance is better understood as robustness to changing sampling geometry rather than transfer across resolutions.

The cylinder flow results also highlight an important practical limitation of one-shot operator learning for unsteady problems. The largest prediction errors occurred in the downstream wake, where some samples exhibited oscillatory behavior associated with transient wake dynamics or vortex shedding. In these cases, Geo-FNO accurately predicted the mean flow structure but often smoothed the instantaneous wake phase. This indicates that part of the remaining error is caused by dataset structure and problem formulation rather than by a failure of the model to learn the dominant physics.

Overall, the results of this thesis show that discretization robustness depends primarily on how a neural operator handles the relationship between continuous functions and their sampled representations. Models that resample inputs onto a fixed internal grid, such as the CNO or FNO variants modified to include this, generalize more reliably across discretizations than models that operate directly on resolution-dependent spectral coefficients. For irregular domains, the same principle remains central: reliable operator learning depends not only on model capacity, but on how consistently the architecture represents the underlying continuous problem across different discretizations.

6.2 Future Work

Several natural extensions of this work remain for future study.

A first priority is the use of more challenging non-uniform reactor datasets. While the HTGR temperature results demonstrate that Geo-FNO can be applied to reactor-relevant data on irregular meshes, the problem itself is relatively simple: the geometry is fixed across all samples, the input consists only of boundary temperature and axial position, and the resulting temperature field is smooth and strongly structured. This makes the experiment closer to a proof of applicability than a demonstration of performance on realistic reactor simulations. Future work should therefore consider datasets involving stronger nonlinear

coupling and more complex physical behavior, such as transient thermal-hydraulics, coupled neutronics and heat transfer, spatially varying material properties, or changing geometries caused by fuel movement or control rod insertion. Such datasets would provide a more meaningful test of whether neural operators can function as practical surrogate models in reactor analysis.

A second important extension is a more systematic study of discretization invariance on non-uniform meshes. In this thesis, no direct resolution-transfer experiments were performed for mesh-free datasets because there is no single notion of resolution for point clouds. Prediction quality depends not only on the number of points, but also on where those points are placed relative to boundaries, sharp gradients, and other important physical features. Future work should therefore construct controlled families of point clouds where local density is varied in a physically meaningful way, for example by refining sampling near cylinder boundaries, wake regions, or thermal boundary layers while keeping the global geometry fixed. This would make it possible to evaluate robustness to sampling geometry directly rather than relying only on held-out prediction quality.

Another limitation of the mesh-free experiments is that only one architecture, Geo-FNO, was evaluated. For the uniform-grid experiments, the contrast between the globally biased FNO and the more locally biased CNO provided an important comparison. For non-uniform meshes, no equivalent comparison was made. This is particularly relevant for the cylinder flow dataset, where the dominant dynamics are strongly local and a local architecture may perform better than Geo-FNO. Future work should therefore compare Geo-FNO against graph-based neural operators or other locally structured mesh-free models, as well as against the common baseline of interpolating the data onto a uniform Cartesian grid and applying a standard FNO. Such comparisons would help determine whether the advantages of Geo-FNO come primarily from learning directly on the original point cloud or from the spectral structure of the architecture itself.

A particularly interesting architectural extension would be the development of a geometry-aware Convolutional Neural Operator (Geo-CNO). One possible approach would be to combine the learned coordinate transformation ϕ^{-1} from Geo-FNO with the CNO architecture by first mapping the irregular point cloud to a more regular computational representation and then applying a CNO instead of an FNO. This would preserve the local inductive bias of convolutional operators while also benefiting from fixed internal resampling and explicit control of bandlimiting. Such an architecture could help determine whether the strong discretization robustness of CNO on uniform grids can be extended to irregular domains and whether locality can improve performance on datasets such as cylinder flow, where Geo-FNO tends to struggle most in the wake region. This direction was also suggested in the original CNO paper and would provide a natural continuation of the work in this thesis.

The cylinder flow results also suggest that future work should investigate arbitrary-time prediction rather than only one-shot prediction of a fixed final state. The largest prediction errors occurred in samples with oscillatory wake behavior, where Geo-FNO often predicted the correct mean wake location but smoothed out the instantaneous downstream oscillations. This indicates that the model captures the dominant flow structure but struggles to predict the exact phase of the unsteady wake. A natural extension would therefore be autoregressive rollout, where the model predicts short time increments and repeatedly advances the

6. CONCLUSIONS AND FUTURE WORK

solution forward in time rather than predicting the distant final state directly. Techniques such as teacher forcing, recurrent rollout stabilization, or hybrid one-shot and autoregressive schemes could make these predictions more stable. This would likely be particularly beneficial for transient fluid problems where phase accuracy matters more than prediction of the mean flow alone.

For uniform-grid neural operators, the modified FNO experiments show that fixed internal resampling is the dominant factor behind improved discretization robustness. This suggests that future work should focus on architectures that combine the efficiency of Fourier-based methods with explicit control of internal representations. Hybrid models that retain spectral convolution while enforcing a stable internal discretization may provide a better balance between accuracy and robustness than standard FNO.

Additional spectral methods could also have strengthened the comparison on uniform-grid problems. A recent example is the Walsh–Hadamard Neural Operator (WHNO) [3], which is better suited to discontinuities and shocks than standard Fourier-based operators. In particular, WHNO–FNO ensembles are shown to outperform both individual models on datasets containing sharp discontinuities. This could be especially relevant for problems such as Darcy flow, where the binary permeability field introduces strong material interfaces. Unfortunately, this work was published after much of the FNO experimentation in this thesis had already been completed and was therefore not included in the experimental comparison.

Similarly, improved anti-aliasing strategies and more accurate filtering schemes could help preserve the benefits of the UND block without unnecessarily attenuating the highest retained modes.

Overall, the central challenge remains the same: constructing neural operators whose internal representations remain faithful to the underlying continuous problem across changing discretizations. Improving this consistency, on both structured and unstructured domains, remains a key direction for making neural operators reliable tools for scientific computing and reactor modeling.

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Appendix A

Disclosure on AI use

AI-based tools were used during this thesis project as support for coding, debugging, troubleshooting, source finding, and comprehension of technical material. During the writing process, they were also used to help structure sections and improve phrasing.

All AI-generated suggestions were reviewed and edited before being included. The research decisions, experiments, interpretation of results, and final thesis text remain my own responsibility.

Appendix B

Navier-Stokes data generation details

This appendix provides additional details on the generation of the self-produced Navier-Stokes dataset used in the uniform grid experiments. The dataset is formulated in vorticity form on the unit square and is generated numerically from the incompressible Navier-Stokes equation [2, 6]:

$$\partial_t \omega + u \cdot \nabla \omega = \nu \Delta \omega + f, \quad (\text{B.1})$$

where ω denotes the vorticity field, u the incompressible velocity field, ν the viscosity, and f an external forcing term. The velocity field is not part of the stored dataset or the prediction target. Instead, it is reconstructed from the vorticity during data generation in order to evaluate the nonlinear advection term, using the streamfunction relation [2, 6]:

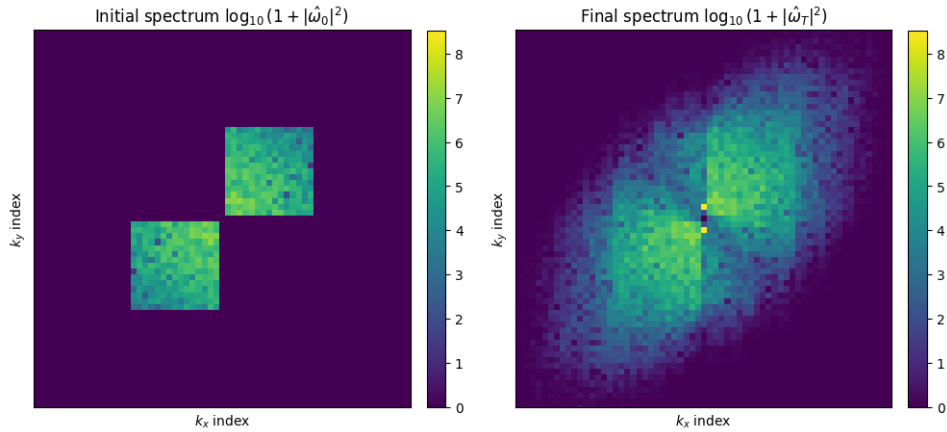


Figure B.1: Fourier spectra of an example initial and final vorticity field from the generated Navier-Stokes dataset. The initial spectrum reflects the truncated low-frequency initialization with Gaussian weighting, while the final spectrum shows the redistribution of spectral energy over time.

$$-\Delta\psi = \omega, \quad u = \nabla^\perp \psi. \quad (\text{B.2})$$

In all generated samples, the viscosity is kept fixed at $\nu = 10^{-5}$. The forcing term is also shared across samples and is given by:

$$f(y, t) = 0.5(1 + 0.5 \sin(0.5t)) \sin(4\pi y), \quad (\text{B.3})$$

which is consistent with the Kolmogorov-flow literature, where the forcing is commonly chosen as a single sinusoidal mode that varies only in one spatial direction [4, 22]. In the present dataset, the spatial factor $\sin(4\pi y)$ serves as the corresponding single-mode forcing on the unit square. The prefactor was chosen so that the forcing remains smooth, bounded, and of moderate magnitude throughout the simulation: specifically, $0.5(1 + 0.5 \sin(0.5t)) \in [0.25, 0.75]$. The additional temporal modulation is not intended to reproduce a specific benchmark forcing law, but was introduced as a synthetic extension to generate trajectories with mild time-dependent variation in the energy input while avoiding abrupt or high-frequency changes.

The dataset contains 1000 samples at a resolution of 256×256 . For each sample, an initial vorticity field is generated in Fourier space. Fourier coefficients are assigned for indices $(k_x, k_y) = (i, j)$ with $i, j \in \{1, \dots, k_{\max} - 1\}$ and $k_{\max} = 16$. These coefficients are random complex numbers whose amplitudes are modulated by a Gaussian envelope of the form $\exp\left(-\frac{1}{2} \frac{i^2 + j^2}{(k_{\max}/2)^2}\right)$, which suppresses higher frequencies and yields smooth initial conditions. The coefficients in the $-k_x, -k_y$ quadrant are then set by conjugate symmetry so that the inverse Fourier transform produces a real-valued field. No coefficients are assigned in the mixed-sign quadrants. The initial vorticity field is finally obtained by applying an inverse Fourier transform, followed by normalization such that the maximum absolute value of the vorticity is 1.

Each initial condition is then evolved forward in time using a fourth-order Runge-Kutta (RK4) integration scheme to obtain the corresponding later vorticity field stored in the dataset.

This initialization procedure produces smooth but diverse initial conditions while concentrating most spectral energy in the low-frequency region. Figure B.1 illustrates this by showing the Fourier spectra of an example initial state and its corresponding final state. The figure highlights both the initially active modes and the effect of the Gaussian envelope, as well as the redistribution of spectral energy under the Navier-Stokes dynamics.

Appendix C

HTGR input engineering details

This appendix provides additional details on the input engineering used for the HTGR column temperature dataset, as introduced in Chapter 3. In this dataset, the model input consists only of the axial level z and the boundary temperature T . Since this input is low-dimensional, several alternative representations were tested to determine whether a more expressive encoding of the same information would improve predictive performance.

Table C.1 shows a representative subset of the tested input representations. The raw representation (z, T) already performs reasonably well, with a test relative RMSE of 3.75×10^{-3} . Replacing these raw variables with min–max normalized variables,

$$\hat{z} = \frac{z - z_{\min}}{z_{\max} - z_{\min}}, \quad \hat{T} = \frac{T - T_{\min}}{T_{\max} - T_{\min}},$$

scales both inputs to the range $[0, 1]$. This improves the test relative RMSE to 2.57×10^{-3} and reduces the test MSE by approximately a factor of two. Thus, for this dataset, giving the model inputs on a common bounded scale is beneficial.

The next representation augments the min–max normalized variables with quadratic and interaction terms, giving $(\hat{z}, \hat{T}, \hat{z}^2, \hat{T}^2, \hat{z}\hat{T})$. This further reduces the test relative RMSE to 2.44×10^{-3} , although the improvement over the basic min–max normalized representation is modest. This is consistent with the fact that $\hat{z}^2$ and $\hat{T}^2$ are not physically meaningful variables for this problem. Instead, they provide a simple polynomial expansion of the two scalar inputs, making some nonlinear dependencies easier for the model to represent.

Finally, a trigonometric representation was tested using the min–max normalized variables together with boundary-aware functions of the axial coordinate: $(\hat{z}, \hat{T}, \sin(\pi(z - z_{\min})/L_z), \cos(\pi(z - z_{\min})/L_z), \hat{z}\hat{T})$, where $L_z = z_{\max} - z_{\min}$. This choice is more directly motivated by the structure of the data. In particular, the sine term vanishes at the axial boundaries, which reflects the observed trend that the peak temperature varies smoothly with z . The cosine term provides a complementary axial feature, and the interaction term $\hat{z}\hat{T}$ gives the model direct access to coupled dependence on axial position and boundary temperature.

The trigonometric representation gives the best performance among the tested configurations, with a test relative RMSE of 2.07×10^{-3} . This shows that using physical understanding of the dataset in the input representation can improve the Geo-FNO prediction

C. HTGR INPUT ENGINEERING DETAILS

Table C.1: Performance comparison of selected input representations for the HTGR column temperature dataset.

Input representation	Val relRMSE	Test relRMSE / MSE
(z, T)	3.3346×10^{-3}	$3.7529 \times 10^{-3} / 7.1206$
min-max normalized $(\hat{z}, \hat{T})$	3.1240×10^{-3}	$2.5735 \times 10^{-3} / 3.4681$
min-max normalized $(\hat{z}, \hat{T}, \hat{z}^2, \hat{T}^2, \hat{z}\hat{T})$	2.9259×10^{-3}	$2.4431 \times 10^{-3} / 2.9970$
$(\hat{z}, \hat{T}, \sin(\pi\hat{z}), \cos(\pi\hat{z}), \hat{z}\hat{T})$	2.1059×10^{-3}	$2.0741 \times 10^{-3} / 2.1801$

accuracy. The trigonometric representation was therefore used for the HTGR results reported in the main text.

Appendix D

Hyperparameters and Training

This appendix provides a complete list of the hyperparameters used for each model and dataset in the experiments described in Chapter 3. The listed hyperparameters were determined through a combination of prior literature, preliminary experiments, and manual tuning to achieve good performance.

For the training of all models, the number of epochs was dictated by a patience scheme. When the validation loss did not improve by at least 0.1% for 25 consecutive epochs, training was stopped. This means that the exact number of epochs varied across models and datasets. The maximum number of epochs was set sufficiently high so that the patience criterion was always met before reaching it, so between 400 and 1000 epochs depending on the model. All models used the Adam optimizer with a cosine annealing learning rate scheduler. The specific learning rates and other hyperparameters are listed in Tables D.1 and D.2.

Some model-specific hyperparameters require brief clarification. For the FNO and Geo-FNO, *modes* (or *modes1* and *modes2*) denote the number of retained Fourier modes in each spatial direction after spectral truncation, while *width* denotes the number of feature channels in the lifted latent representation. For the CNO, *N_layers* controls the depth of the encoder-decoder structure, *N_res* and *N_res_neck* denote the number of residual blocks in the encoder/decoder and bottleneck (lowest layer) respectively, and *channel_multiplier* determines the number of feature channels at each spatial resolution.

For Geo-FNO, the parameters *s1* and *s2* define the size of the internal uniform grid used after the initial non-uniform Fourier transform and coordinate mapping. The first Fourier layer operates directly on the irregular input point cloud after the learned coordinate transform ϕ^{-1} , while the remaining layers use this fixed internal grid. For the cylinder flow dataset, the batch size was set to 1 because different samples contain different numbers of spatial points depending on the cylinder geometry. Constructing minibatches with variable point counts requires additional padding or masking strategies, and batch size 1 provided the simplest and most stable training setup.

D. HYPERPARAMETERS AND TRAINING

Dataset	Model	batch_size	learning rate	modes	width
Navier-Stokes	FNO_k8	4	2×10^{-4}	8	96
	FNO_k16	4	2×10^{-4}	16	128
Darcy flow	FNO_k8	4	2×10^{-4}	8	96
	FNO_k16	4	2×10^{-4}	16	128

Table D.1: The used hyperparameters for the FNO models. Only modes is listed, as modes1 = modes2 = modes was used for all models.

Dataset	Model	batch_size	learning rate	N_layers	N_res	N_res_neck	channel_multiplier
Navier-Stokes	CNO_standard	10	5×10^{-4}	4	4	4	16
	CNO_no_UND	10	5×10^{-4}	4	4	4	16
Darcy Flow	CNO_standard	10	5×10^{-4}	4	4	4	16
	CNO_no_UND	10	5×10^{-4}	4	4	4	16

Table D.2: The used hyperparameters for the CNO models.

Dataset	batch_size	learning rate FNO	learning rate IPHI	modes1	modes2	width	s1	s2
Cylinder flow	1	1×10^{-3}	1×10^{-4}	24	12	32	100	25
HTGR temperature	8	1×10^{-3}	5×10^{-5}	16	16	32	64	64

Table D.3: The used hyperparameters for the Geo-FNO models.

Appendix E

Geo-FNO Bug Fix

During testing of the Geo-FNO, a persistent vertical seam-like artifact appeared across all tested datasets. The artifact was particularly visible in the cylinder flow dataset, where predictions showed a clear discontinuity along the center of the domain. An example is shown in Figure E.1, where the predicted solution exhibits a sharp artificial transition that is not present in the ground truth.

Inspection of the learned coordinate transformation ϕ^{-1} revealed that it consistently split the domain along the same vertical line, as illustrated in Figure E.2. At first glance, this suggested that the learned coordinate mapping itself might be responsible for the artifact. However, training the model without ϕ^{-1} caused the seam to become substantially more pronounced, while the prediction quality away from the seam remained largely unchanged

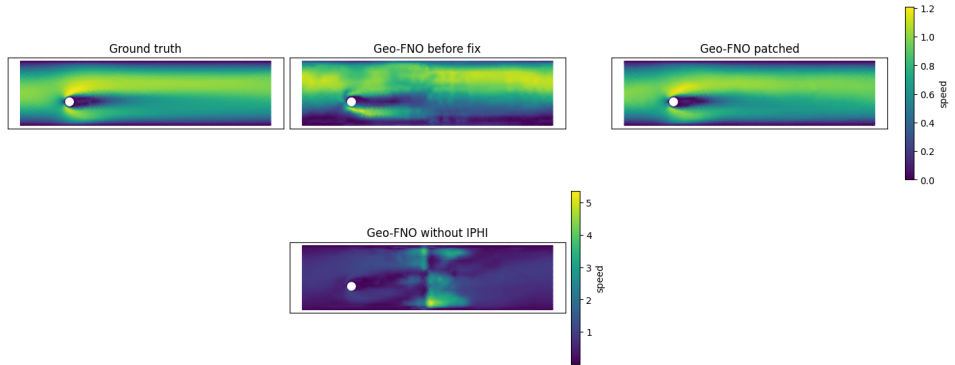


Figure E.1: Representative Geo-FNO prediction on the cylinder flow dataset before and after correcting the Fourier mode reconstruction bug. In the first row, from left to right, the ground truth, the original prediction containing the seam artifact, and the corrected prediction after patching the negative-mode reconstruction are shown using a shared color scale for direct comparison. The second row shows the prediction of the unpatched model trained without the learned coordinate mapping ϕ^{-1} . This case is displayed with a separate colorbar due to the significantly larger artifact amplitude and resulting difference in value range.

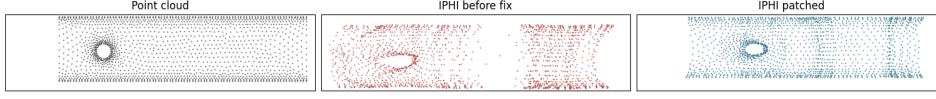


Figure E.2: Learned coordinate mapping ϕ^{-1} before and after the bug fix. The left panel shows the input grid before any coordinate mapping ϕ^{-1} has been applied. The middle panel shows the mapping learned by the original implementation, which exhibits a vertical split of the domain. The right panel shows the mapping learned after correcting the Fourier mode reconstruction, which no longer exhibits this artificial split.

and the overall loss increased only modestly. This indicates that the learned behavior of ϕ^{-1} was not the source of the artifact, but rather an adaptation to it. By reducing the density of sampled points near the corrupted region, the mapping partially mitigated the visible effect of the reconstruction error without removing its underlying cause.

The source of the artifact was found to be an error in the reconstruction of discarded Fourier modes during the inverse transform. As in the standard FNO, Geo-FNO retains only a finite number of Fourier modes in each direction. For a real-valued field, the Fourier coefficients satisfy the Hermitian symmetry condition

$$U(-k_x, -k_y) = \overline{U(k_x, k_y)}, \quad (\text{E.1})$$

where the overline denotes complex conjugation. This symmetry implies that only half of the Fourier spectrum needs to be stored explicitly, since the remaining coefficients can be reconstructed from their conjugate counterparts. In principle, any conjugate half of the spectrum could be discarded; in the Geo-FNO implementation, the modes with negative k_y are omitted for memory efficiency and reconstructed during the inverse transform using Equation E.1.

The implementation keeps only the modes with positive k_y , while separate learned complex weights are applied to the $(+k_x, +k_y)$ and $(-k_x, +k_y)$ branches. During reconstruction of the physical field, the discarded negative k_y modes are recovered using Equation E.1. The seam artifact was caused by an indexing error in this reconstruction step.

The original implementation reconstructed the discarded modes using a simple reversal of the stored mode ordering. For the k_y direction, this is correct. However, for the k_x direction, the Fourier modes are stored in the order

$$[0, 1, \dots, N-1, -N, -(N-1), \dots, -1]. \quad (\text{E.2})$$

Two modes in this ordering are self-conjugate: the zero mode $k_x = 0$ and the Nyquist edge mode $k_x = -N$. Applying a direct reversal produces

$$[-1, -2, \dots, -N, N-1, N-2, \dots, 0], \quad (\text{E.3})$$

which incorrectly pairs the modes during reconstruction.

As a result, the implemented reconstruction effectively enforced

$$U(k_x, k_y) = \overline{U(-(k_x + 1), -k_y)}, \quad k_x > 0, \quad (\text{E.4})$$

instead of the correct Hermitian symmetry relation in Equation E.1. This mismatch was strongest for the lowest k_x modes adjacent to the self-conjugate modes, causing corruption of the low-frequency components of the reconstructed field. Since these low modes dominate the global structure of the solution, the error appeared visually as a strong vertical seam through the center of the domain.

Correcting the reconstruction so that the discarded modes were matched to their proper conjugate partners removed the artifact entirely. For the cylinder flow dataset, this reduced the mean squared error from approximately 5.5×10^{-3} to 4.7×10^{-3} , corresponding to a reduction of approximately 15%. The visual seam disappeared completely, as shown in Figure E.1, while the learned ϕ^{-1} mapping no longer exhibited the artificial domain split shown in Figure E.2.

The issue was reported to the authors of the original Geo-FNO implementation, and the proposed correction was subsequently incorporated into the official repository [25].